Lecture 07 Investment and Arbitrage

Part 1 - Synthetics and Derivatives

Derivatives in Traditional Finance (TradFi)

- Derivatives are financial instruments whose price/payouts are determined from the underlying asset
- Value is “derived” from another asset
- Derivatives examples:
  - Futures
  - Options
  - Swaps

Understanding Derivatives: The Big Picture

How derivatives derive their value from underlying assets



Key Insight: Derivatives allow risk transfer without owning the asset

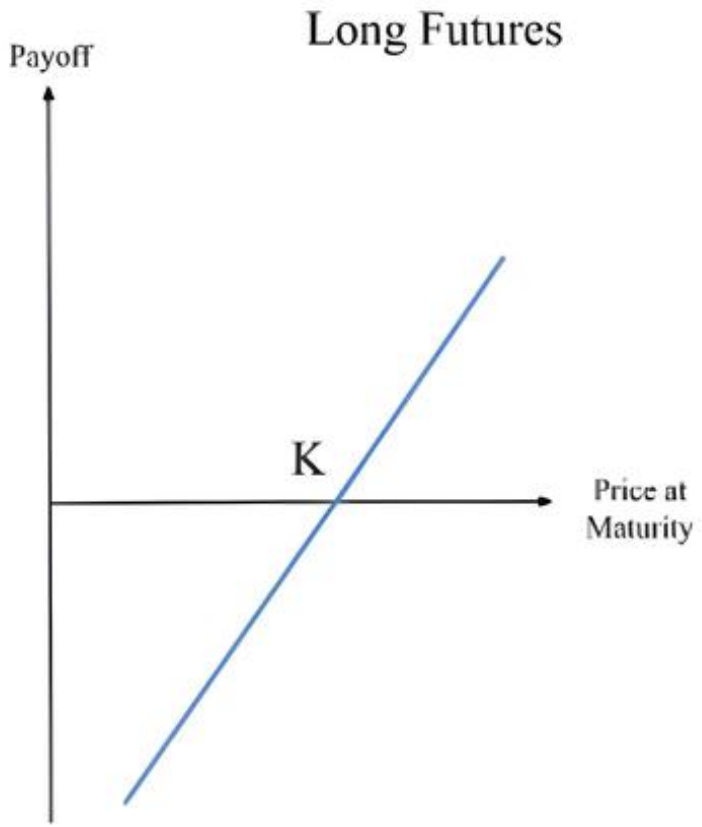
Global derivatives market: ~$700 trillion notional value (2024)

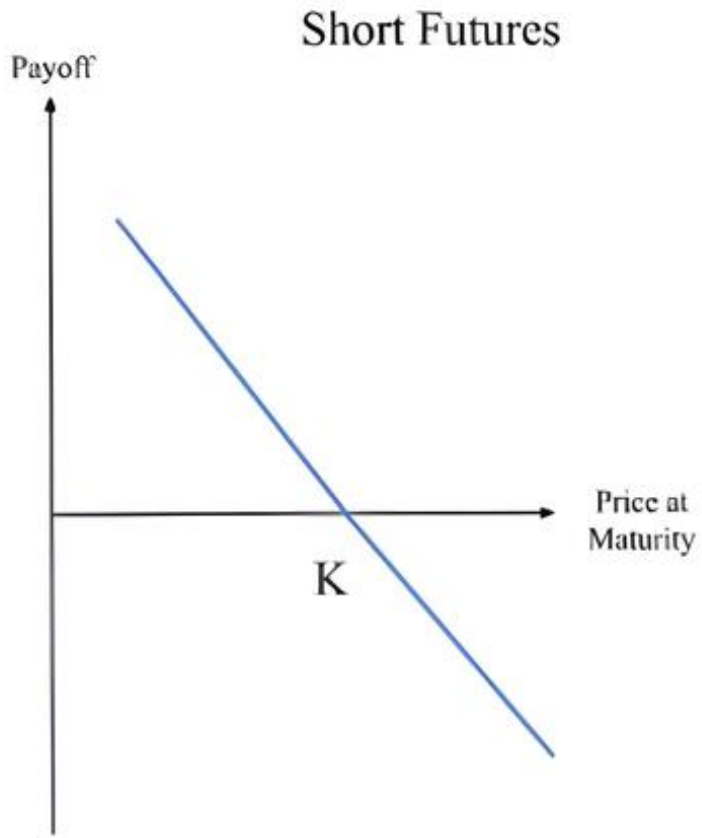
Much larger than the stock market (~$100T) or bond market (~$130T)

Futures: Buy Now, Pay Later

- A contract to agree to buy an asset in the future at a fixed price
- Settlement in cash or underlying
- Example:
  - Trader – farmer contract: Agrees to buy corn at the price of $10/kg in 3 months
  - After 3 months: Price of corn = $11
  - Trader profits $1, Farmers guaranteed a predictable income
- Counterparty risk: What if trader disappears?
- Futures are standardized and written by central parties – reducing counterparty risk and making them fungible

Futures: Return Profiles





Futures in Practice: Oil Futures Example

Scenario Setup

An airline wants to lock in fuel costs.
Current oil price: $70/barrel
Futures contract: Buy at $72 in 6 months
Contract size: 1,000 barrels

If oil rises to $85/barrel:
Airline saves: ($85 - $72) x 1,000 = $13,000

If oil drops to $60/barrel:
Airline overpays: ($72 - $60) x 1,000 = $12,000
But gained price certainty for budgeting!

Key Concepts

Hedging vs Speculation

Hedgers (like the airline) use futures to reduce risk.
Speculators use futures to profit from price changes.

Marking to Market

Profits/losses settled daily, not at expiry. This reduces counterparty risk.

Margin Requirements

Only need to deposit 5-15% of contract value as margin. This provides leverage.
Example: $72,000 contract needs only ~$7,200 margin = 10x leverage

Why are Futures Contracts Designed Like This?

- Fills a missing economic function
  - Risk transfer from producers to speculators.
  - Producers can lock in price today which makes planning easier.
- Maximizes market liquidity
  - Contract size (dollar value) chosen so most potential participants will find it attractive.
  - Exchange takes care of collatheral management so traders don't have to expend time and effort to find a safe counterparty.
  - Common price due to marking to market makes it easier to trade.

Options

- Options are types of derivatives that give owners a right, but not an obligation, to buy or sell an asset at an agreed price
- Options are written by a market maker and sold to an owner – the market maker is obligated to sell/buy the asset if the owner exercises its right to buy/sell by an expiry date
- Option writer charges a premium for writing the option and taking the risk

Options: Key Terminology

Strike Price

The predetermined price at which the option holder can buy (call) or sell (put) the asset.

Premium

The price paid to purchase the option. This is the maximum loss for the buyer.

Expiry Date

The deadline by which the option must be exercised. After this date, the option expires worthless.

In-The-Money (ITM)

Call: Spot > Strike. Put: Spot < Strike. The option has intrinsic value.

Out-of-The-Money (OTM)

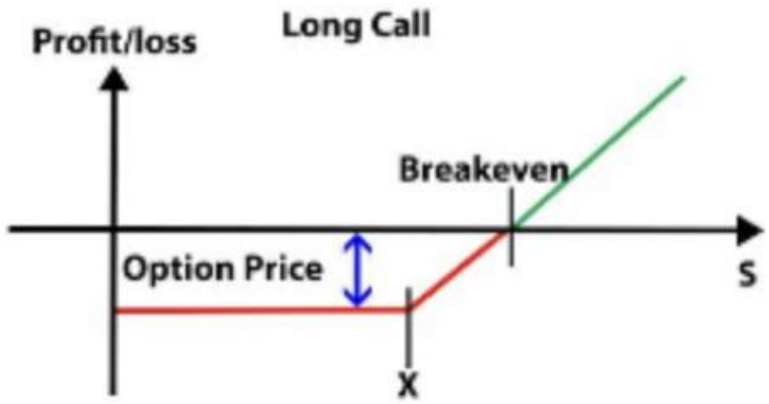
Call: Spot < Strike. Put: Spot > Strike. The option has no intrinsic value.

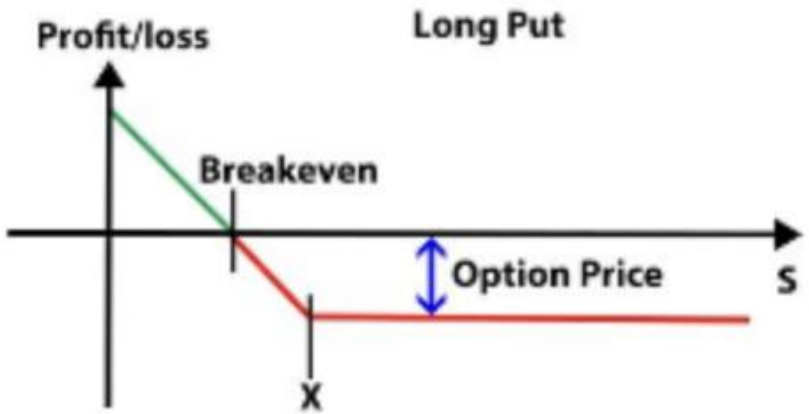
Intrinsic vs Time Value

Option value = Intrinsic value (profit if exercised now) + Time value (possibility of future profit).

Options: Return Profiles

- Call – Option owner has a right to buy asset at an agreed price
- Put – Option owner has a right to sell asset at an agreed price





Options: Concrete Payoff Examples

Call Option Example

Strike Price: $100

Premium Paid: $5

Expiry: 30 days

Scenario A: Price = $120

Profit = ($120 - $100) - $5 = $15

Return: +300%

Scenario B: Price = $90

Option expires worthless

Loss: -$5 (premium only)

Scenario C: Price = $105

Profit = ($105 - $100) - $5 = $0

Break-even point

Put Option Example

Strike Price: $100

Premium Paid: $4

Expiry: 30 days

Scenario A: Price = $80

Profit = ($100 - $80) - $4 = $16

Return: +400%

Scenario B: Price = $110

Option expires worthless

Loss: -$4 (premium only)

Scenario C: Price = $96

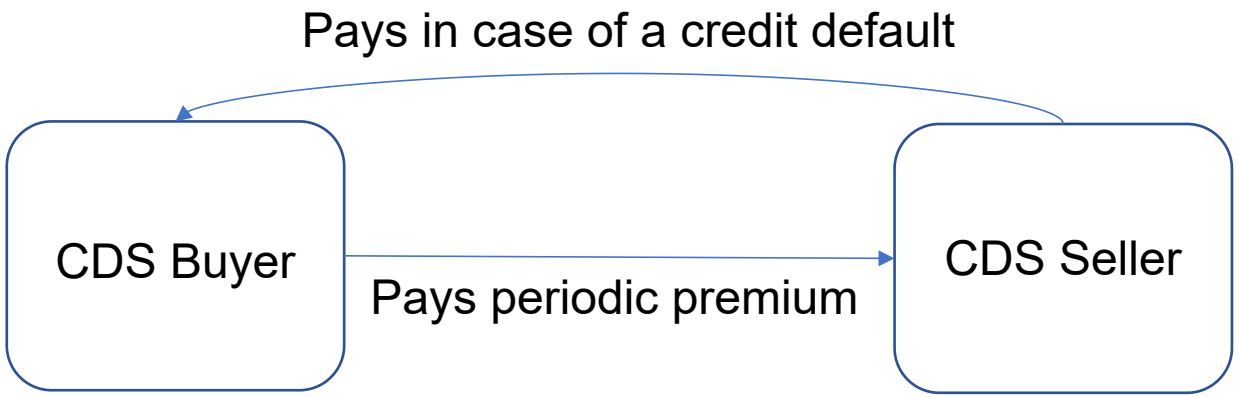
Profit = ($100 - $96) - $4 = $0

Break-even point

Swaps: Bilateral Derivatives

- Exchange asset value without transferring assets
- Example:
  - Credit Default Swaps (Pure trading of credit risk, like insurance)

    (CDS buyer and seller might have absolutely nothing to do with the underlying company whose credit risk they are trading.)



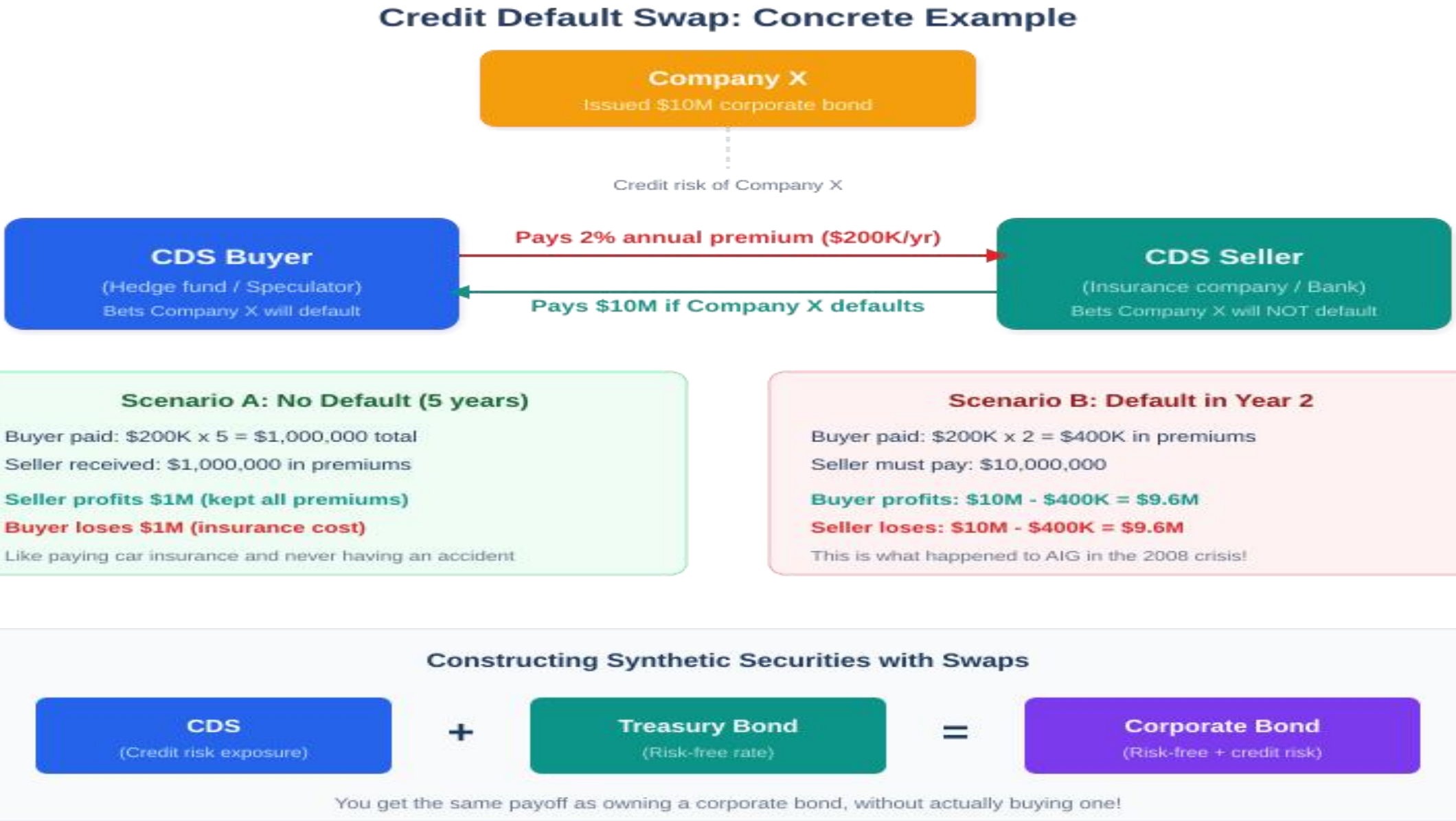
Swaps: Bilateral Derivatives

Why are swaps designed this way?

- Want a specific economic exposure but too expensive to sell + buy underlying.
- Cash flows only exchanged on the notional value ⇨ lower counterparty risk.

- Flexible way to construct synthetic securities, i.e.,
  - credit default swap + treasury = corporate bond

Credit Default Swap: Real-World Example



DeFi Synthetic Assets: Tokenized Derivatives

- Tokenization of assets
  - Cryptocurrency: WBTC, WETH
  - Stock: Apple, Tesla
  - Commodity: gas, gold
  - Index: DeFi Pulse Index
- Futures
  - Perpetuals: DyDx
- Options
  - Ribbon



Types of Tokenization of Assets

1. Wrapping cryptocurrencies
   - Actually hold the assets
   - Wrapping a basket of cryptocurrencies (indexing)

2. Tracking asset prices
   - No need to hold the assets
   - Synthetics

Wrapping as A Tokenization Mechanism

- Price is pegged to another asset



Need for Synthetic Assets

- Bring real world assets to blockchain – Synthetic Gold, USD pegged Stablecoins
- Remove need for token transfers across chains - WBTC
- Enable complex financial instruments like futures and options
- Borderless access to assets – Apple stock on-chain

TradFi vs DeFi: Derivatives Comparison

Feature	Traditional Finance	DeFi
Intermediary	Banks, exchanges, clearinghouses	Smart contracts (no intermediary)
Access	Accredited investors, KYC required	Permissionless, anyone with a wallet
Settlement	T+2 days typically	Instant (within one block, ~12 sec)
Trading Hours	Market hours (9:30 AM - 4 PM)	24/7/365
Transparency	Limited, opaque OTC markets	Fully transparent on-chain
Counterparty Risk	Managed by clearinghouse	Managed by over-collateralization
Custody	Held by broker/custodian	Self-custody (your keys, your assets)
Composability	Limited integration	Fully composable ("DeFi Legos")

Real World Assets On-Chain

- Representing real world assets on-chain relies on:
  - Maintaining asset in real world reserves
  - Enabling price discovery using oracles
  - Mint/burn mechanism to deposit/redeem asset from the reserves
- Examples:
  - Centralized Stablecoins: USDC
  - Stocks on chain: Apple
  - On-chain commodities: PAX Gold

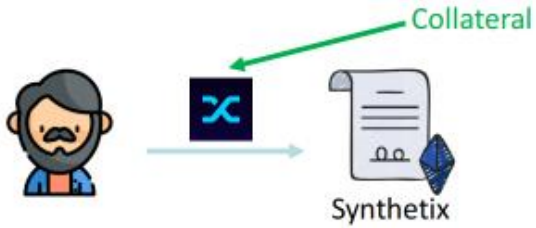
Collateralized Debt Position (CDP) Based Synthetics

- Use collateral asset through Ethereum smart contract to mint a synthetic token called “synths”.
  - Example: Synthetix (token: SNX)
- Utilize oracles to valuate the synthetic
- Maintain a minimum collateral ratio (MCR)

CDP Based Synthetics

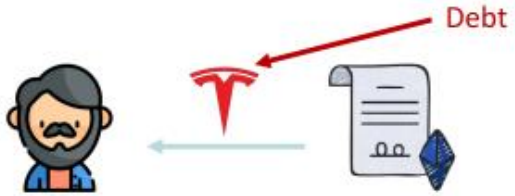
Mint a derivative asset

Step 1



SNX deposited to mint derivative
600% collateral mints 100% derivative

Step 2



Draw derivative
(e.g. sTSLA share)

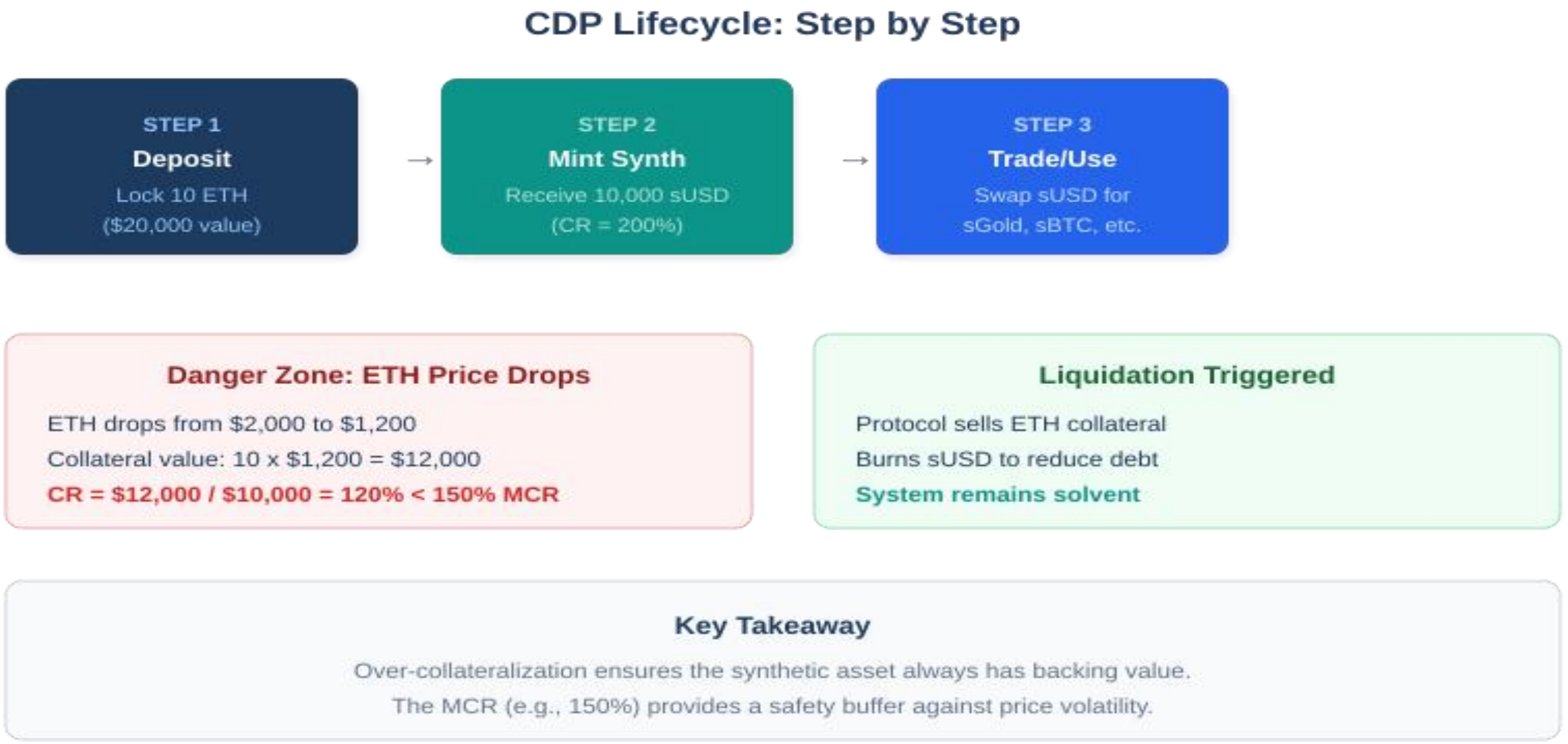


Step 3



Pay back sTSLA to unlock SNX

CDP Lifecycle: Step-by-Step Process



CDP Based Synthetics

- If value of synthetic asset rises, liquidation pressure leads to buying pressure – decreasing supply of the synthetic

Assumptions	
Minimum Collateral Ratio	150%
Collateral Auction Discount	0%
Initial Shares of Asset	1.00
Initial Asset Price	$90
Initial Collateral	$150

When the collateral ratio falls below the minimum collateral ratio, the protocol will sell collateral to buy back shares of the minted asset to burn

Time	Shares of Asset	Asset Price	Collateral Value	Collateral Ratio	
0	1.000	$90	$150	167%	(=$90 / $150)
1	1.000	$100	$150	150%	
2	1.000	$101	$150	149%	<150%
2.01	0.970	$101	$147	150%	
3	0.970	$110	$147	137%	<150%
3.01	0.727	$110	$120	150%	
4	0.727	$100	$120	165%	

Many Different Synths are Offered

- Fiat Currency Synths
  - sUSD, sEUR, sKRW.
- Commodity Synths
  - synthetic gold and synthetic silver, both measured per ounce;
- Cryptocurrencies
  - sBTC, sETH.
- Also offer inverse Synths e.g. iBTC.
  - If the price of the underlying does down, these go up in value
  - Same payoffs as a **short** position in TradFi
- Also Synths that track indexes (similar to ETF)

How to Ensuring Price Pegs for CDP Based Synthetics?

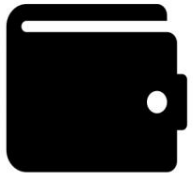
1. Arbitrage: If the price of a synth is too low, stakers can profit from buying sUSD back below par and burning it to reduce their debt.
2. sETH liquidity pool on Uniswap: a portion of SNX added to the total supply (due to inflationary policy) is distributed as a reward to sETH/ETH liquidity providers.
3. SNX auction: partnering with dFusion protocol, discount SNXs are sold for ETH, which is then used to purchase synthetic assets below peg.

CDP Based Synthetics

- Features
  - Can be generalized to any derivative given oracle and liquidator support
  - Pools are not needed for exchanges:
    - Convert from one synthetic to another according to price oracle
    - Zero slippage, infinite liquidity – there can be more sBTC than the total supply of BTC
- Vulnerabilities
  - As secure as the price of the collateral – Collapse of Mirror on Terra blockchain
  - Reliance on oracles

CDP Based Synthetics Benefit:

Infinite Liquidity "Buying" sBTC with sUSD



- Charge a fee
- Remove (burn) sUSD.
- Add sBTC at a rate determined by the oracle



- System converts the debt from one Synth to another.
- No counterparty needed
- Infinite liquidity
- No Debt change is required as Synth pool increases and decreases by the same amount

Futures On-Chain

- Tracking asset prices
  - No need to hold the assets
  - Perpetuals (contract without an expiry date)
  - Total cumulative perpetual trading volume since 2020 has surpassed 60 Trillion USDs
  - For context, in 2019 the total global stock market trading volume was around 60 Trillion.

Perpetual Futures: A Concrete Example

Example: Trading BTC-PERP on dYdX

Setup: Alice deposits $1,000 USDC as margin on dYdX. She opens a 10x leveraged Long BTC-PERP position.

Position size: $10,000 worth of BTC | Entry price: $60,000/BTC | Margin: $1,000 (10x leverage) | No expiry date

BTC rises to $66,000

Price change: +10%

Position P&L: +$1,000

(= $10,000 x 10%)

Return on margin: +100%

10x leverage amplifies the 10% price move to 100% profit

Balance: $1,000 + $1,000 = $2,000

BTC drops to $57,000

Price change: -5%

Position P&L: -$500

(= $10,000 x -5%)

Return on margin: -50%

Only 5% price drop but Alice already lost half her margin

Balance: $1,000 - $500 = $500

BTC drops to $54,000

Price change: -10%

Position P&L: -$1,000

(= $10,000 x -10%)

LIQUIDATED!

Loss = entire margin. Protocol closes position automatically.

Balance: $0 (100% loss)

Perp vs Spot: Key Differences

- No actual BTC exchanged (just profit/loss in USDC)
- Can go Long OR Short (profit from price drops too)
- Leverage amplifies both gains AND losses
- No expiry — hold until you close or get liquidated

Don't Forget: Funding Rate Cost

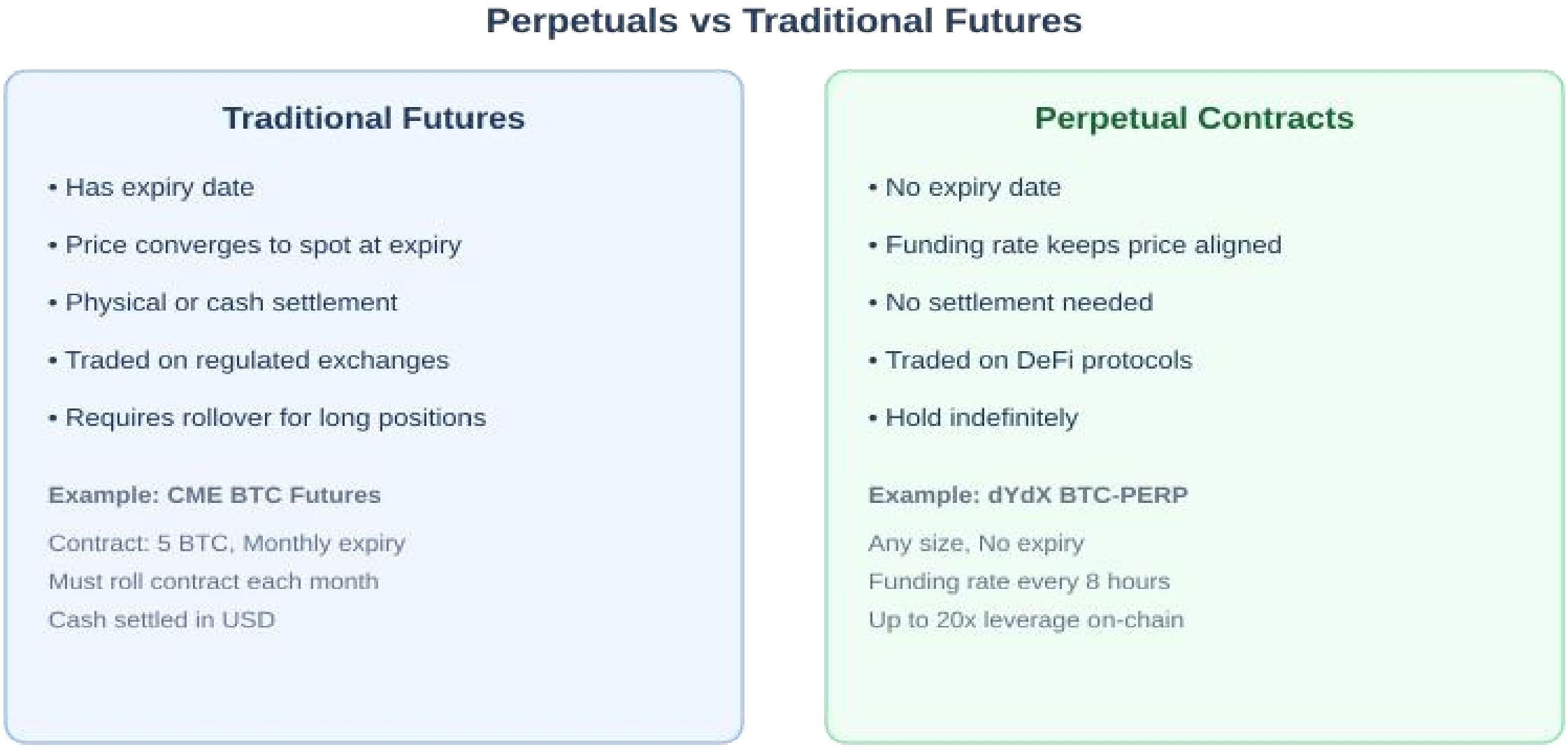
Alice also pays/receives funding rate every 8 hours.

If funding = +0.01%, Alice pays: $10,000 x 0.01% = $1

That's $3/day or ~$90/month — eats into profits!

This is how the perp price stays pegged to spot.

Perpetuals vs Traditional Futures



dYdX: Perpetuals

- Perpetuals speculate on future price of an asset without any expiry date – Futures without expiry
- There is no exchange of assets between buyers and sellers
  - No need to actually exchange assets at expiry
- Short and long positions on the perpetual contract create the market
- Price of perpetual contract is controlled by the funding rate mechanism

Funding Rate: How Perpetuals Stay Pegged

Funding Rate Mechanism Explained

Perp Price > Spot Price

Funding Rate is POSITIVE

Longs pay Shorts

Discourages going long

→ Selling pressure → Price drops to spot

Perp Price < Spot Price

Funding Rate is NEGATIVE

Shorts pay Longs

Discourages going short

→ Buying pressure → Price rises to spot

Numerical Example (BTC-PERP on dYdX)

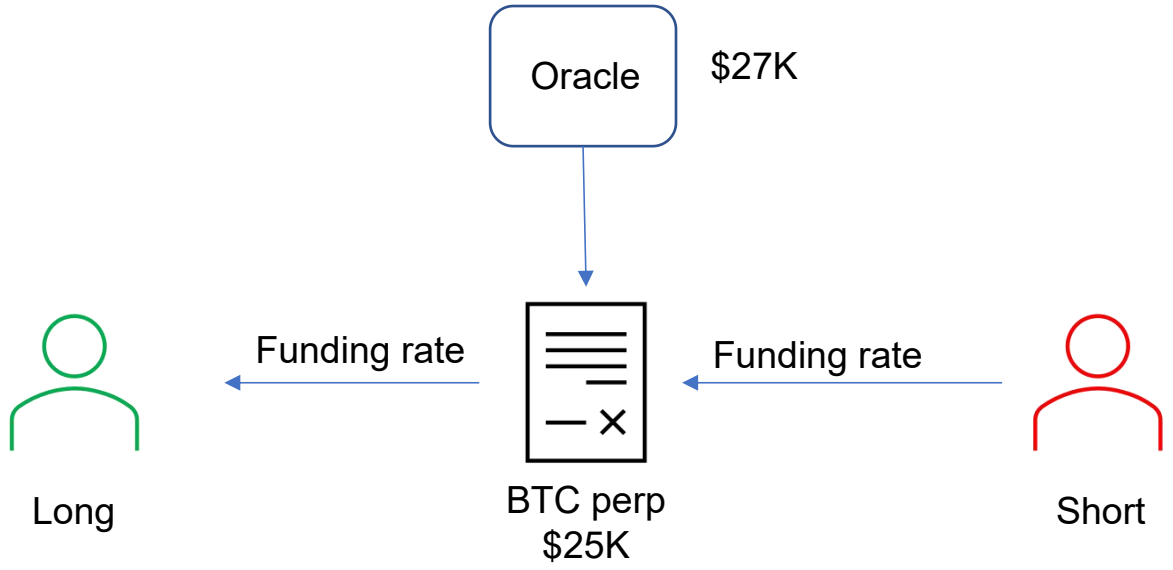
BTC Spot Price: $60,000 | Perp Price: $60,300 | Funding Rate: +0.01%

If you hold a $100,000 long position, you pay: $100,000 x 0.01% = $10 per 8 hours

This cost incentivizes longs to close, reducing buying pressure until Perp = Spot

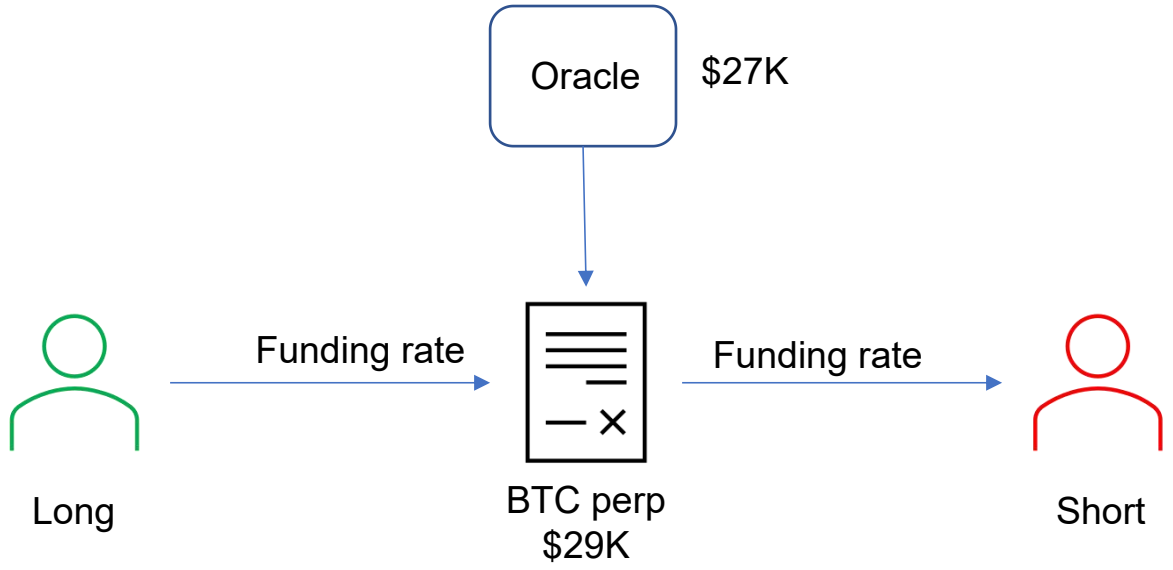
dYdX: Perpetuals – Price Discovery

- Holders of long positions pay holders of short position if price of perpetual contract is lower than the trading price of asset – purchasing pressure



dYdX: Perpetuals – Price Discovery

- Holders of short positions pay holders of long position if price of perpetual contract is higher than the trading price of asset – selling pressure



Funding Rate: Concrete Example

Funding Rate Example: Perp Price ($29K) > Spot Price ($27K)

Current State: Oracle (Spot) = $27,000 | BTC Perp on dYdX = $29,000 | Premium = +$2,000 (+7.4%)

The perp is trading ABOVE the real market price. Funding rate is POSITIVE. Longs must pay shorts.

Alice (Long Position)

Position size: $100,000 (long BTC perp)

Funding rate: +0.05% per 8 hours

Alice PAYS: $100,000 x 0.05% = $50 every 8h

= $150/day = ~$4,500/month

Holding a long position is now EXPENSIVE

Bob (Short Position)

Position size: $100,000 (short BTC perp)

Funding rate: +0.05% per 8 hours

Bob RECEIVES: $100,000 x 0.05% = $50 every 8h

= $150/day = ~$4,500/month

Holding a short position EARNS money

What Happens Next? (Market Self-Correction)

1. Some longs close their positions to avoid paying $150/day → selling pressure → perp price drops
2. New traders open shorts to earn $150/day for free → more selling pressure → perp price drops further

Result: Perp price gradually drops from $29K back toward $27K (spot). Funding rate shrinks toward 0.

Funding Rate Formula (Simplified)

Funding Rate = (Perp Price - Spot Price) / Spot Price

= ($29K - $27K) / $27K = +7.4%

Divided across 8-hour intervals over the period

Basis Trading Strategy

Buy 1 BTC spot ($27K) + Short 1 BTC perp ($29K)

= Delta-neutral + collect funding rate

Professional traders use this for low-risk yield

Options On-Chain

- DeFi option vaults
- AMM LP payoff-derived oracle-free options

DeFi Option Vaults - Ribbon

- Mints options with weekly expiration
- Number of options minted are controlled by the liquidity available to the protocol
- Premium on the option is calculated through an auction mechanism
- Profit/loss as an option maker is transferred to the protocol’s LPs

Ribbon Option Vault: Concrete Example

Ribbon Vault: Weekly Covered Call Example

STEP 1 (Monday) — **LPs Deposit** — 100 ETH into vault

→

STEP 2 (Friday) — **Vault Mints Options** — 100 covered call options

→

STEP 3 (Friday) — **Auction** — Market makers bid

→

STEP 4 (Next Fri) — **Settlement** — Expire or exercise

Concrete Numbers: ETH spot = $2,000 | Vault deposits: 100 ETH | Strike price: $2,200 (10% OTM) | Expiry: 1 week

Auction result: Market maker pays 0.5 ETH premium per contract. Total premium collected: 100 x 0.5 = **50 ETH**

The vault is now a covered call writer: it holds 100 ETH (collateral) and has sold 100 call options at strike $2,200

Scenario A: ETH stays below $2,200

Example: ETH = $2,100 at expiry

Options expire worthless (OTM, no exercise)

LPs keep 100 ETH + 50 ETH premium

Weekly yield: 50/100 = 50% (annualized huge!)

This is the ideal outcome for vault depositors

Scenario B: ETH rises above $2,200

Example: ETH = $2,500 at expiry

Options exercised: must sell ETH at $2,200

LPs miss upside: sold at $2,200 vs $2,500

Lost opportunity: $300/ETH x 100 = $30,000

But still keep the 50 ETH premium as consolation

Why This is Revolutionary

In TradFi: Writing options requires expertise, capital, and a broker account.

In DeFi: Just deposit ETH. Vault does the rest.

LP = Option Writer (Key Insight)

LPs provide collateral = they are the option sellers

Profit/loss flows to LPs, not the protocol

Risk: Capped upside, full downside exposure

Part 1 Recap: Key Takeaways

- Derivatives derive value from underlying assets: Futures, Options, and Swaps
- DeFi tokenizes derivatives: wrapping, CDP-based synthetics, perpetuals, options vaults
- CDPs use over-collateralization + liquidation to maintain synthetic asset pegs
- Perpetuals use funding rates instead of expiry dates to track spot prices
- DeFi composability enables combining protocols like "money Legos"
- Key risk: oracle dependency, smart contract bugs, and collateral volatility

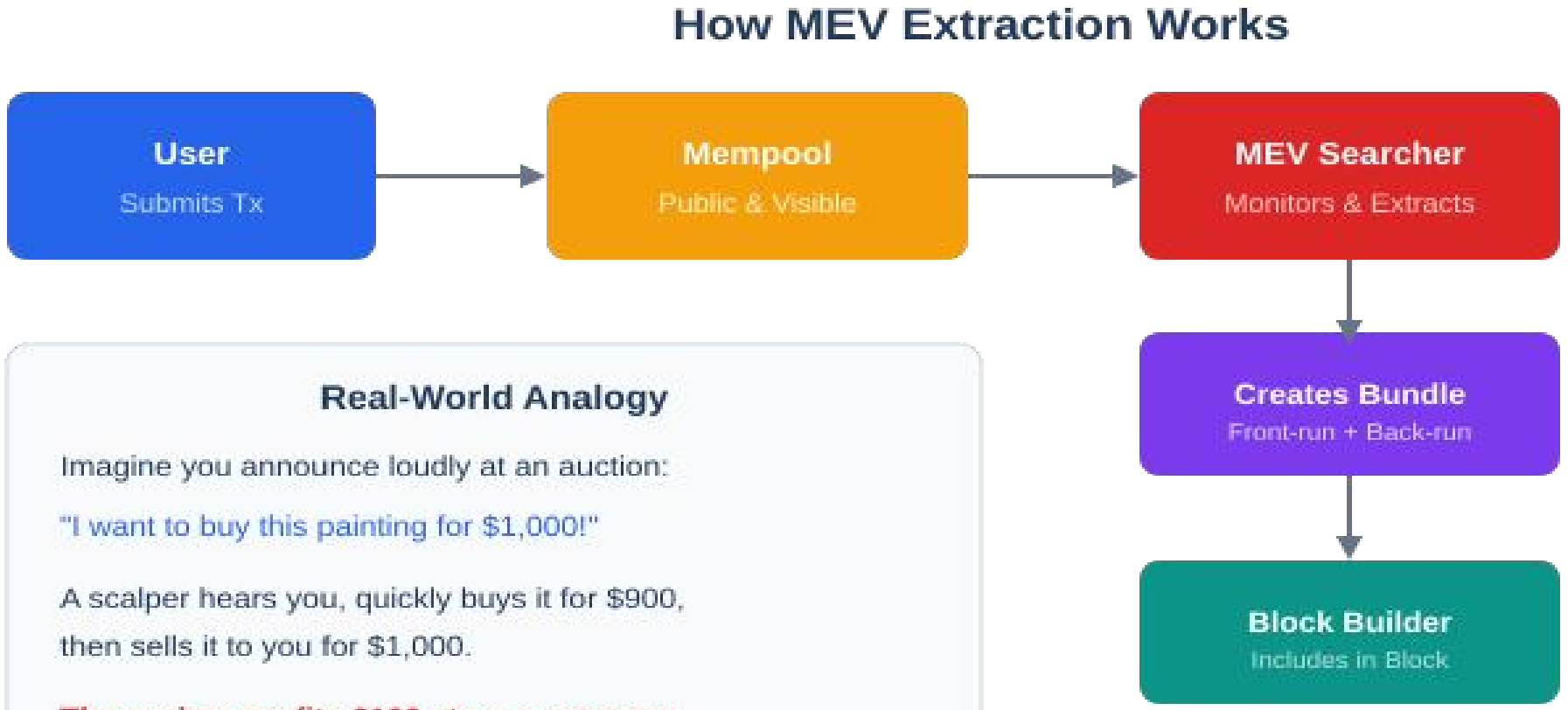
Part2 - Maximal Extractable Value (MEV)

Miner/Maximal Extractable Value (MEV)

Recall - What is MEV?

MEV is a measure of the profit that can be made through including, excluding, or re-ordering transactions within blocks.

MEV: Understanding Through Real-World Analogy



Real-World Analogy

Imagine you announce loudly at an auction:

"I want to buy this painting for $1,000!"

A scalper hears you, quickly buys it for $900, then sells it to you for $1,000.

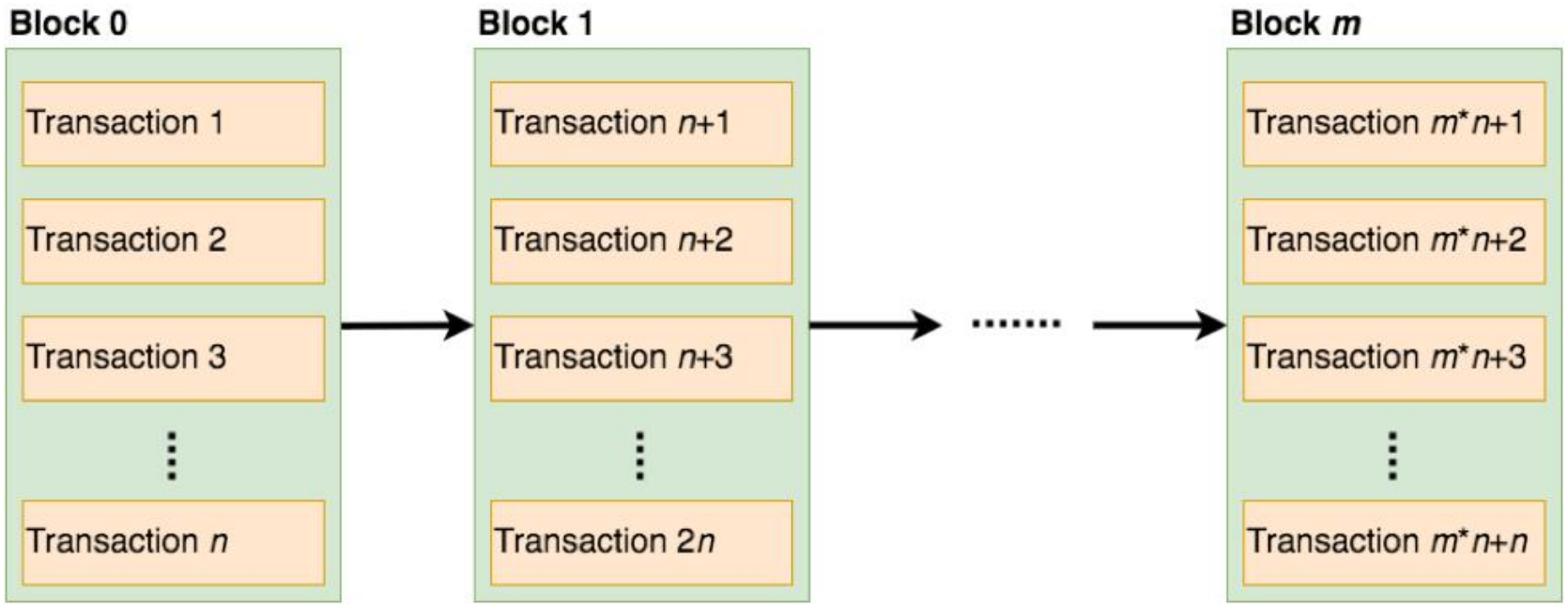
The scalper profits $100 at your expense.

This is essentially how front-running works in MEV!

MEV extracted on Ethereum since 2020: Over $600 million (source: Flashbots)

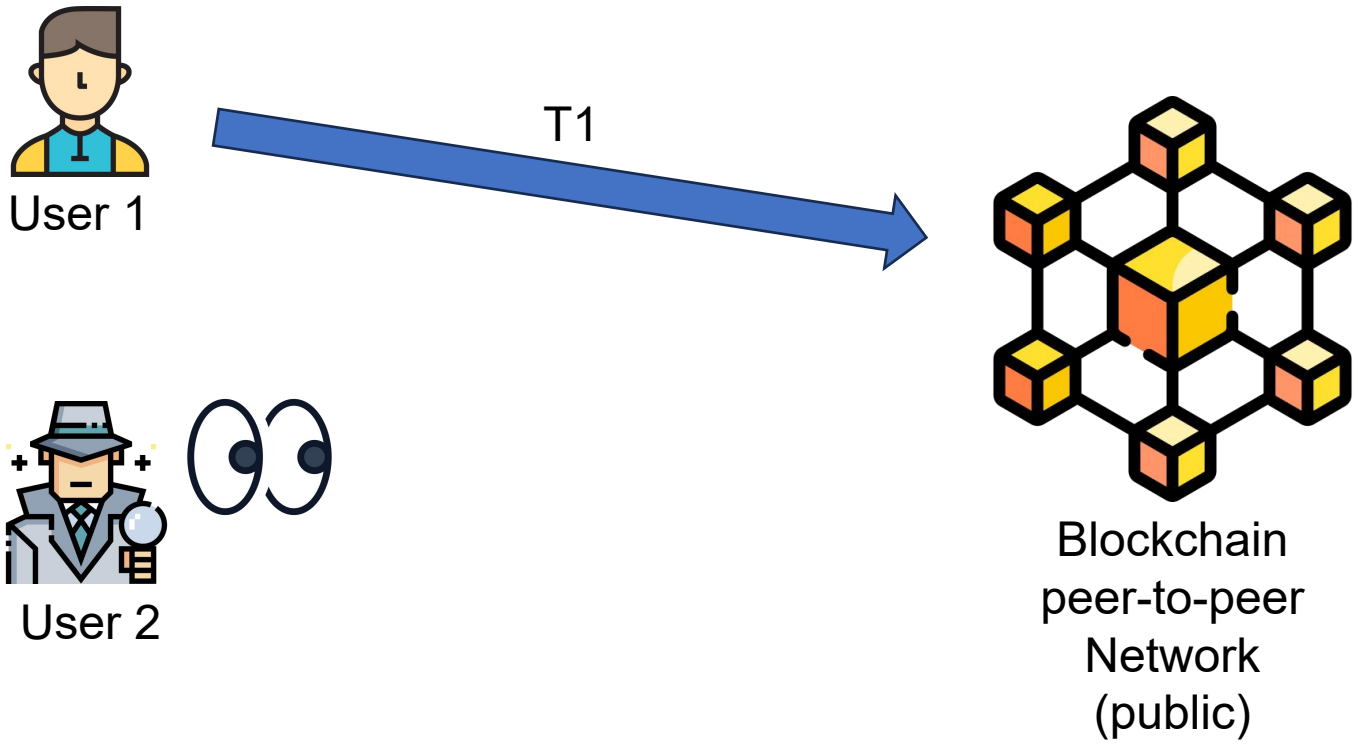
Recall - Why MEV Appear?

1. Transaction is collected and executed in a block, not executed immediately after it sent.



Recall - Why MEV Appear?

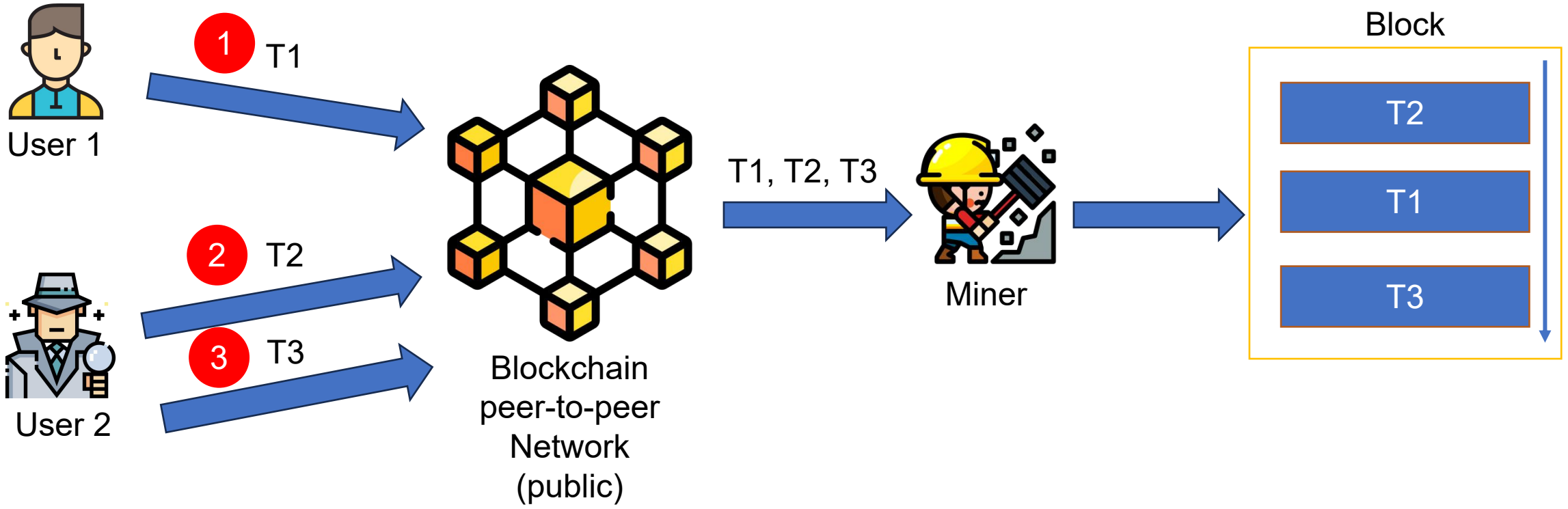
2. All transactions are public in the Blockchain Network (Information Transparency)



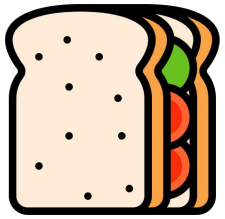
Recall - Why MEV Appear?

3. Later Transaction (T2) can be executed before the earlier transaction (T1).

   It can be happened by providing higher gas price to T2 compared to T1, or to bribe miner in order to change the order of the transactions in the block.



3 Types of MEV Sources



Sandwich Attacks
(In DEX)



Liquidations
(In Lending)



Arbitrage
(In DEX)

Three Types of MEV: Detailed Comparison

Sandwich Attack

- Front-run + back-run victim
- Occurs on DEX trades
- Exploits slippage tolerance
- Victim gets worse price
- Revenue: price difference
- Common on Uniswap, Sushi

Arbitrage

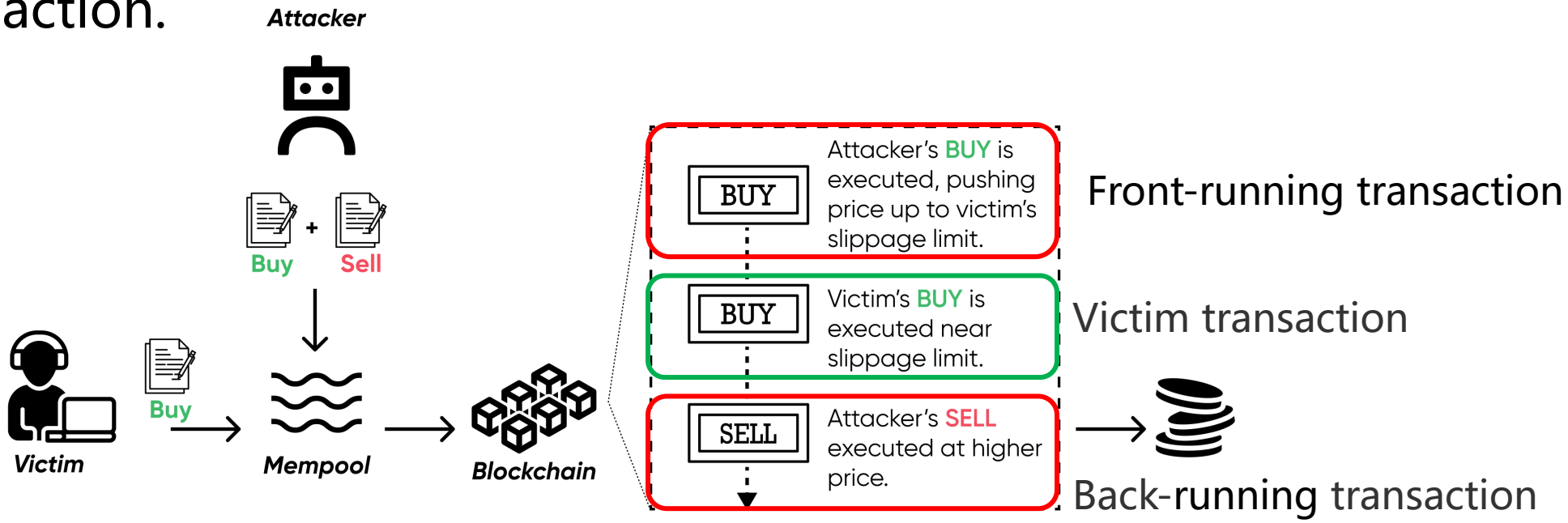
- Cross-DEX price equalization
- Benefits market efficiency
- Requires speed & capital
- No direct victim harmed
- Revenue: price discrepancy
- Most common MEV type

Liquidation

- Repay undercollateralized debt
- Occurs in lending protocols
- Earns liquidation bonus (5-15%)
- Protects protocol solvency
- Revenue: discount on collateral
- Common on Aave, Compound

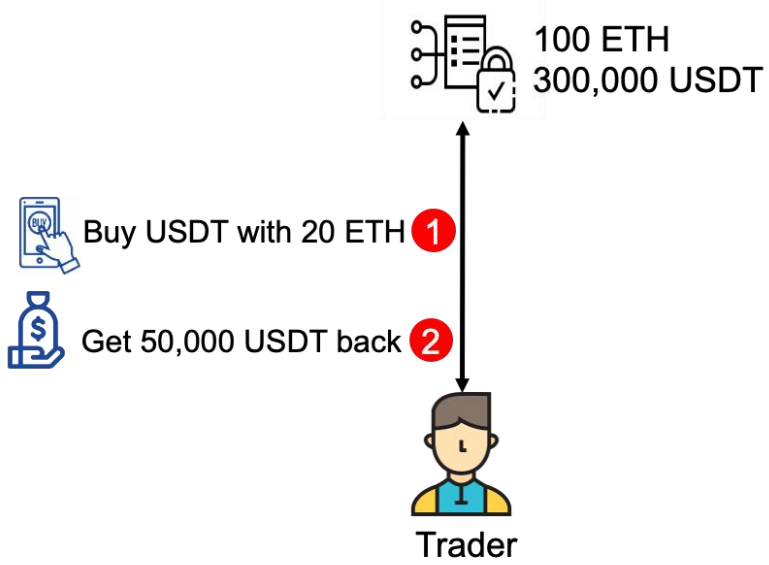
Sandwich Attack

- An attacker observes a non-executed transaction (victim transaction) in the blockchain P2P network. The attacker places one order right before the user's transaction and one right after it, effectively "sandwiching" the user's transaction.



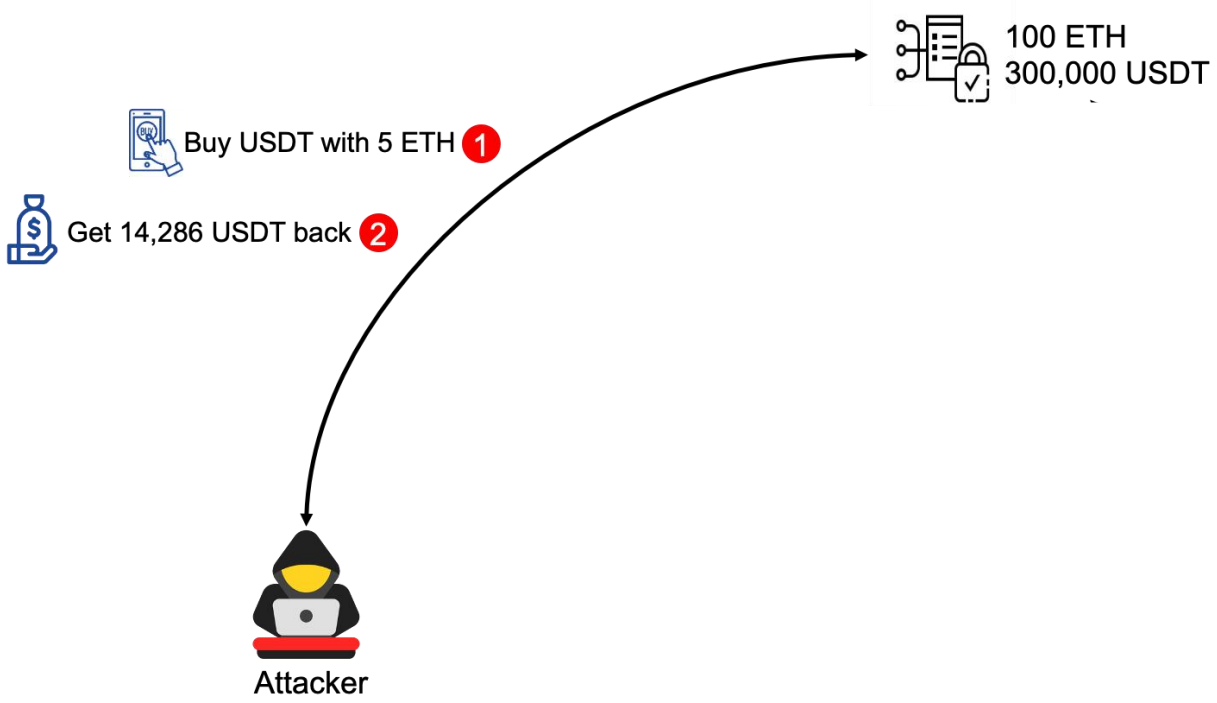
Sandwich Attack Example

- An example of an asset exchange on AMM DEXes with constant product function and no commission fee.
  - The trader sends 20 ETH to the liquidity pool and gets 50,000 USDT back.
  - The product of the amount of ETH and USDT in the liquidity pool is constant, i.e., 30,000,000.



Sandwich Attack Example

- Front-running transaction





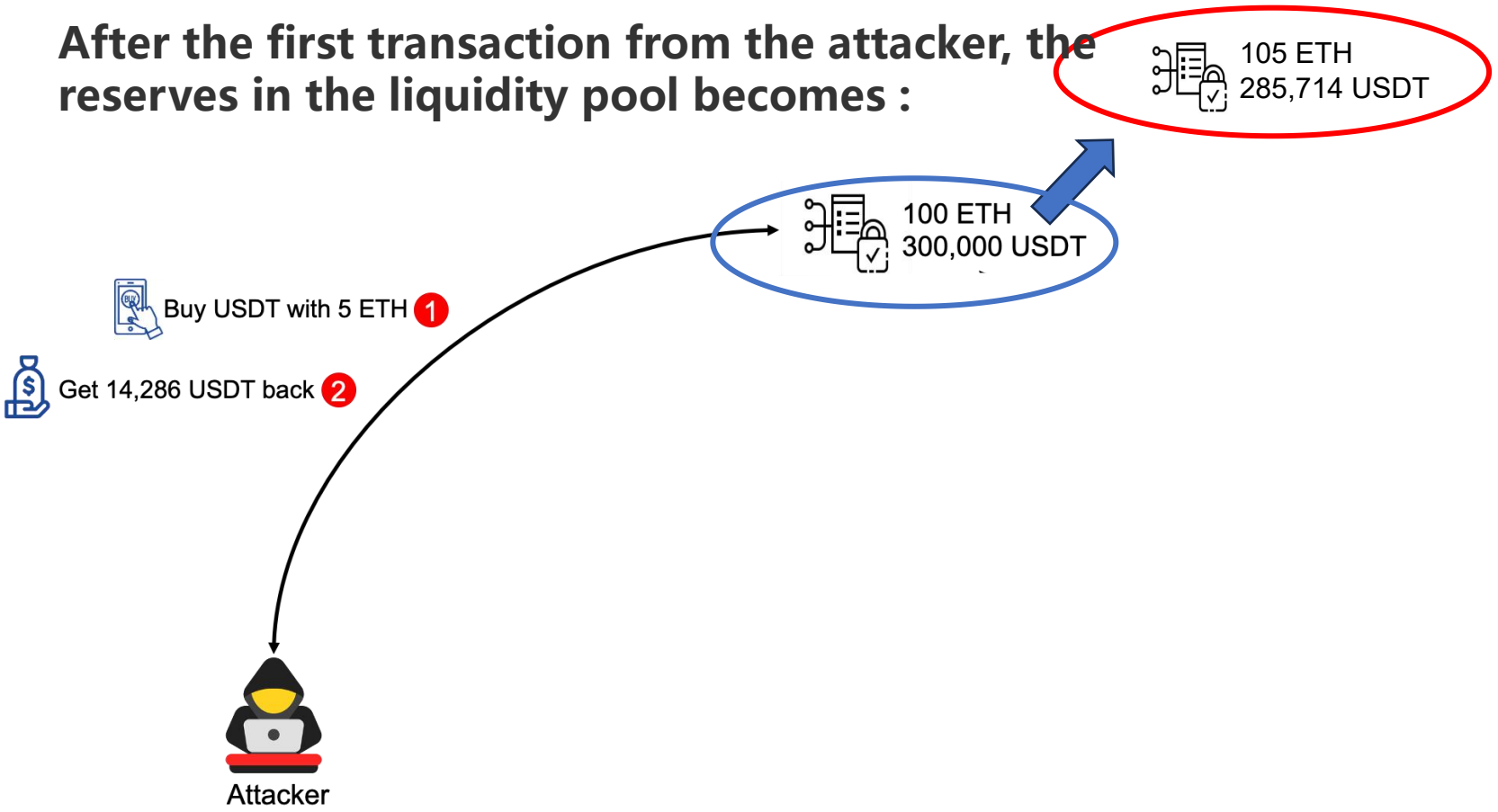
Total:
100*300,000=30,000,000

Step 1:
Attacker buys USDT with 5 ETH
(100+5) * (300,000- X) = 30,000,000

Step 2:
How many USDT can the attacker get?
X = 14,286

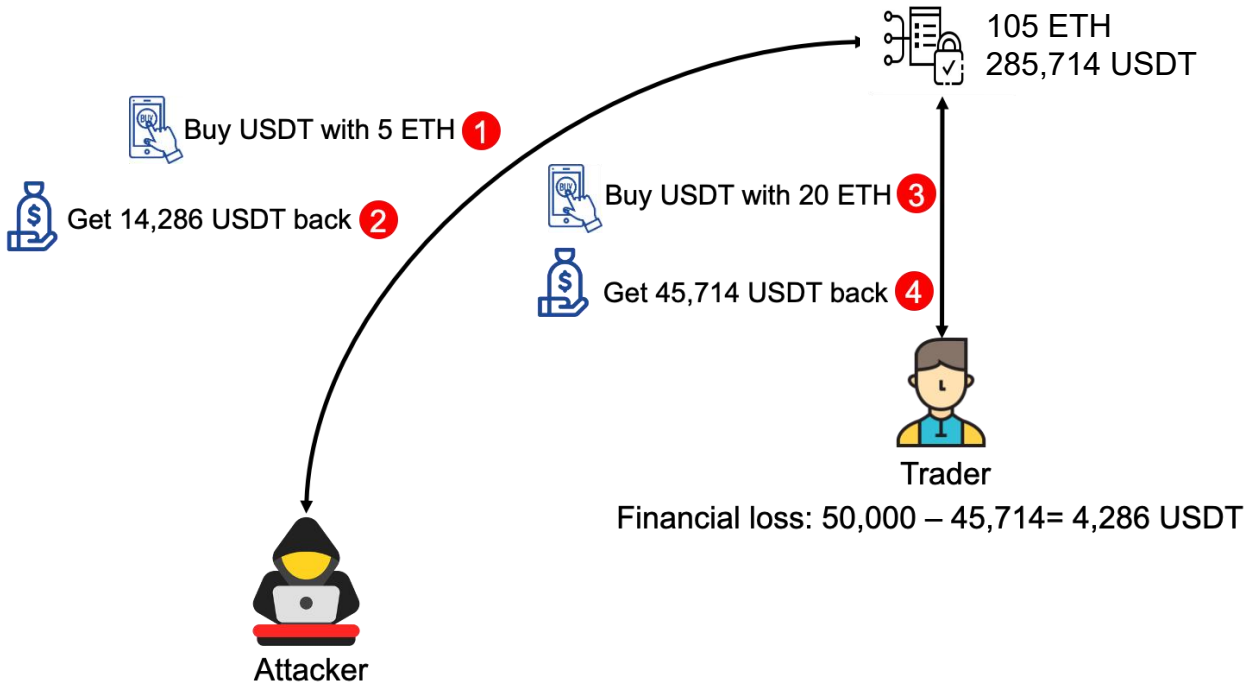
Sandwich Attack Example

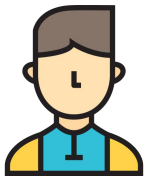
After the first transaction from the attacker, the reserves in the liquidity pool becomes :



Sandwich Attack Example

- Victim transaction

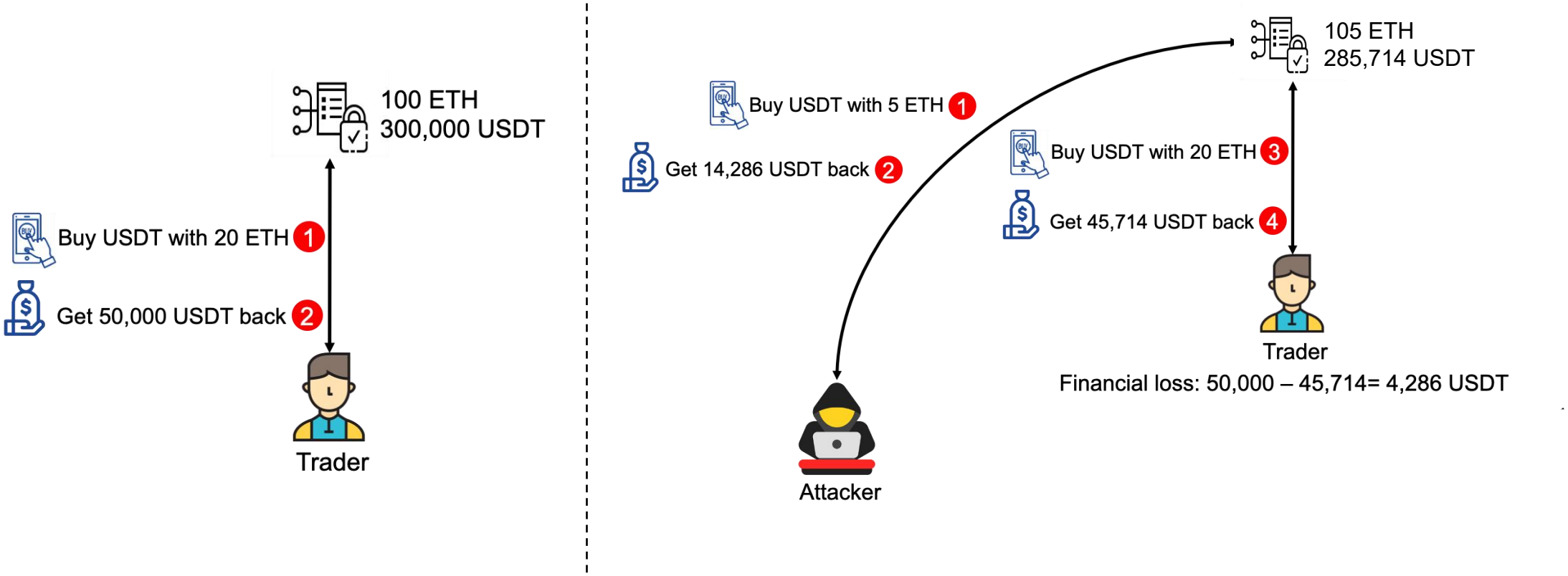




Step 3:
The trader buys USDT with 20 ETH
(105+20) * (285,714- Y) = 30,000,000

Step 4:
How many USDT can the trader get?
Y = 45,714

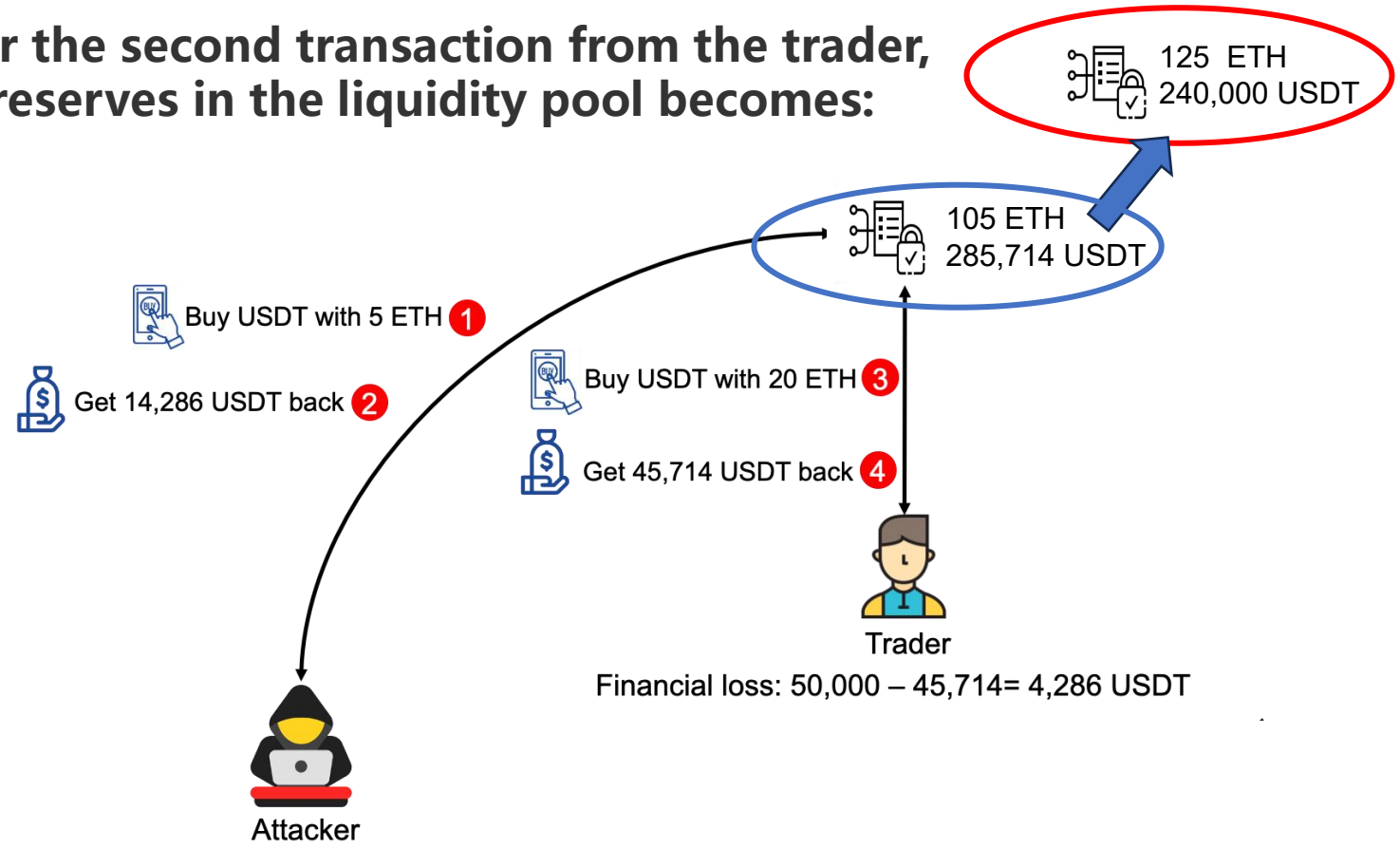
Sandwich Attack Example



The trader has a financial loss with 4,286 USDT when the attacker submits a front-running transaction.

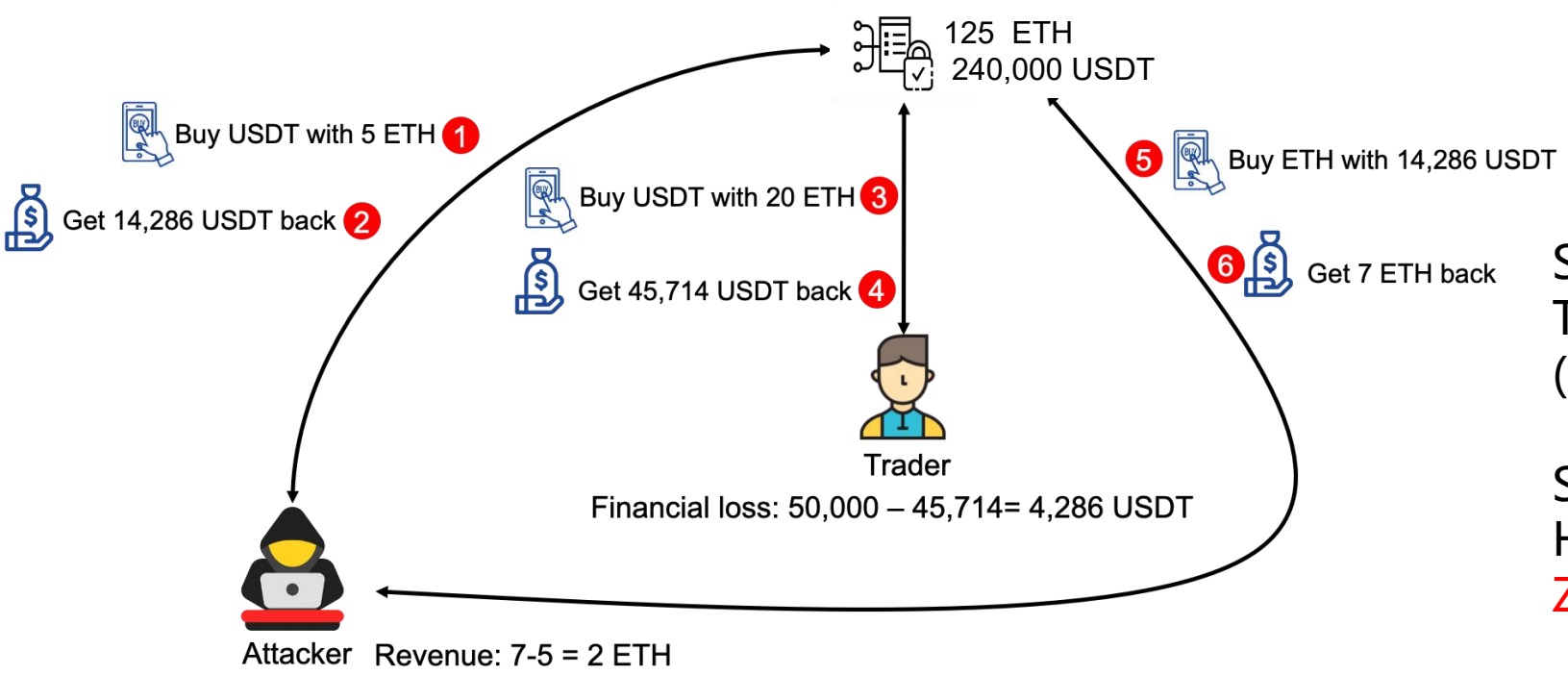
Sandwich Attack Example

After the second transaction from the trader, the reserves in the liquidity pool becomes:



Sandwich Attack Example

- Back-running transaction

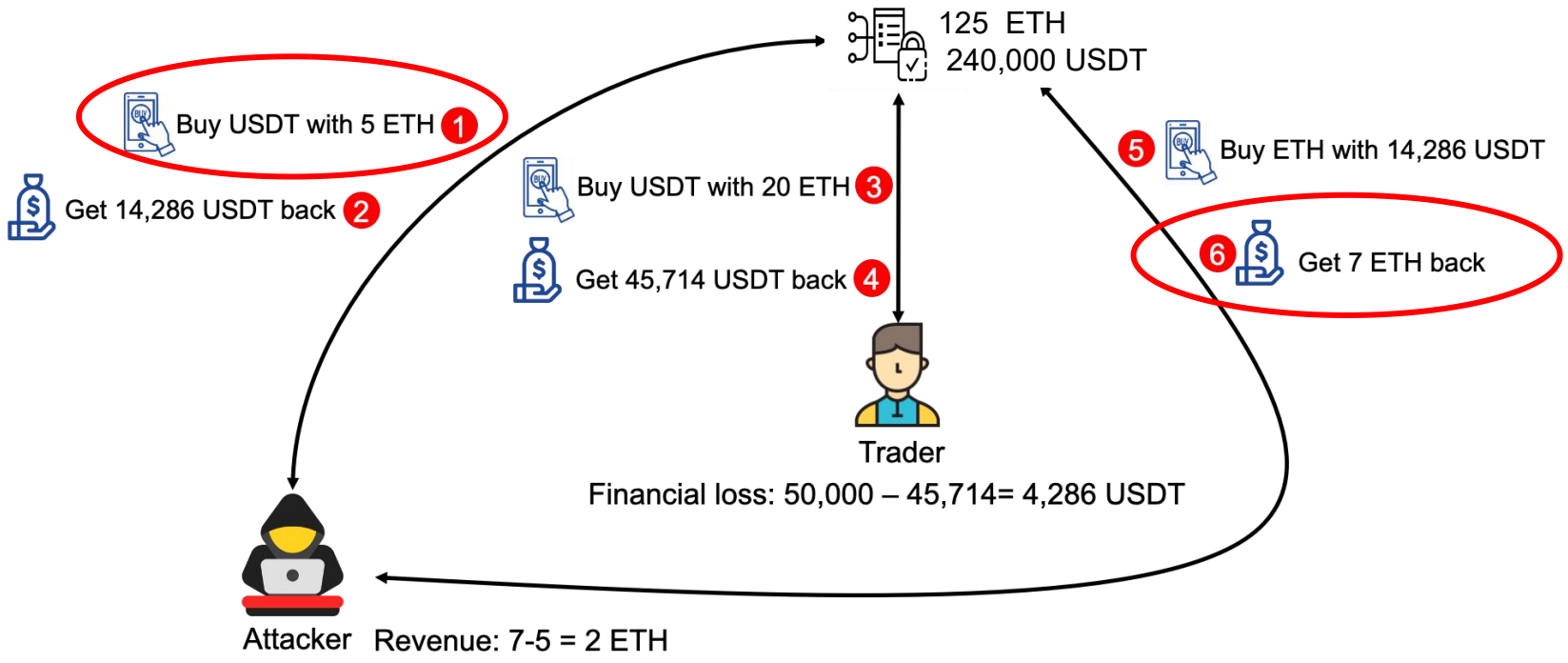




Step 5:
The attacker buys ETH with 14,286 USDT
(125-Z) * (240,000+14,286) = 30,000,000

Step 6:
How many ETH can the attacker get?
Z = 7

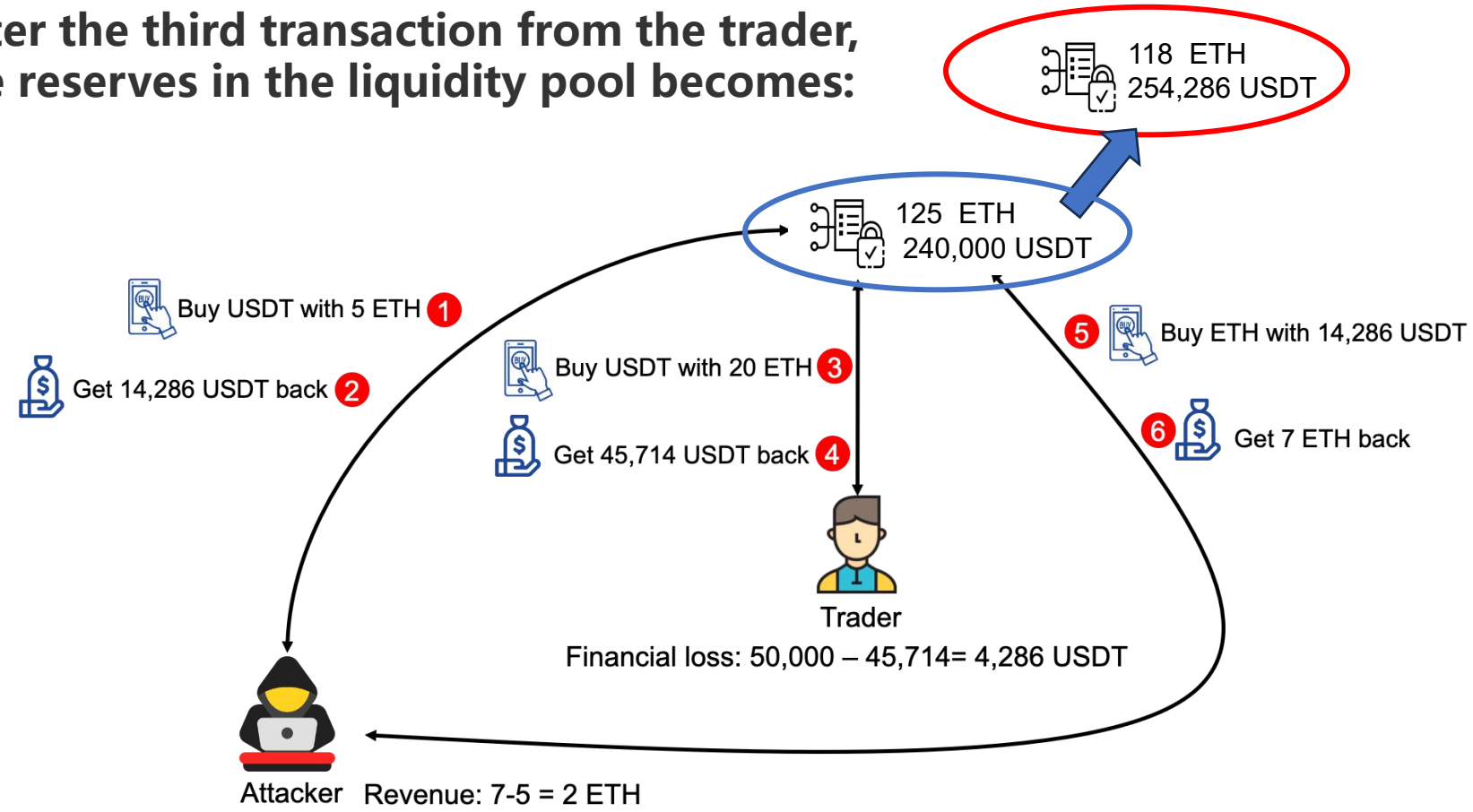
Sandwich Attack Example



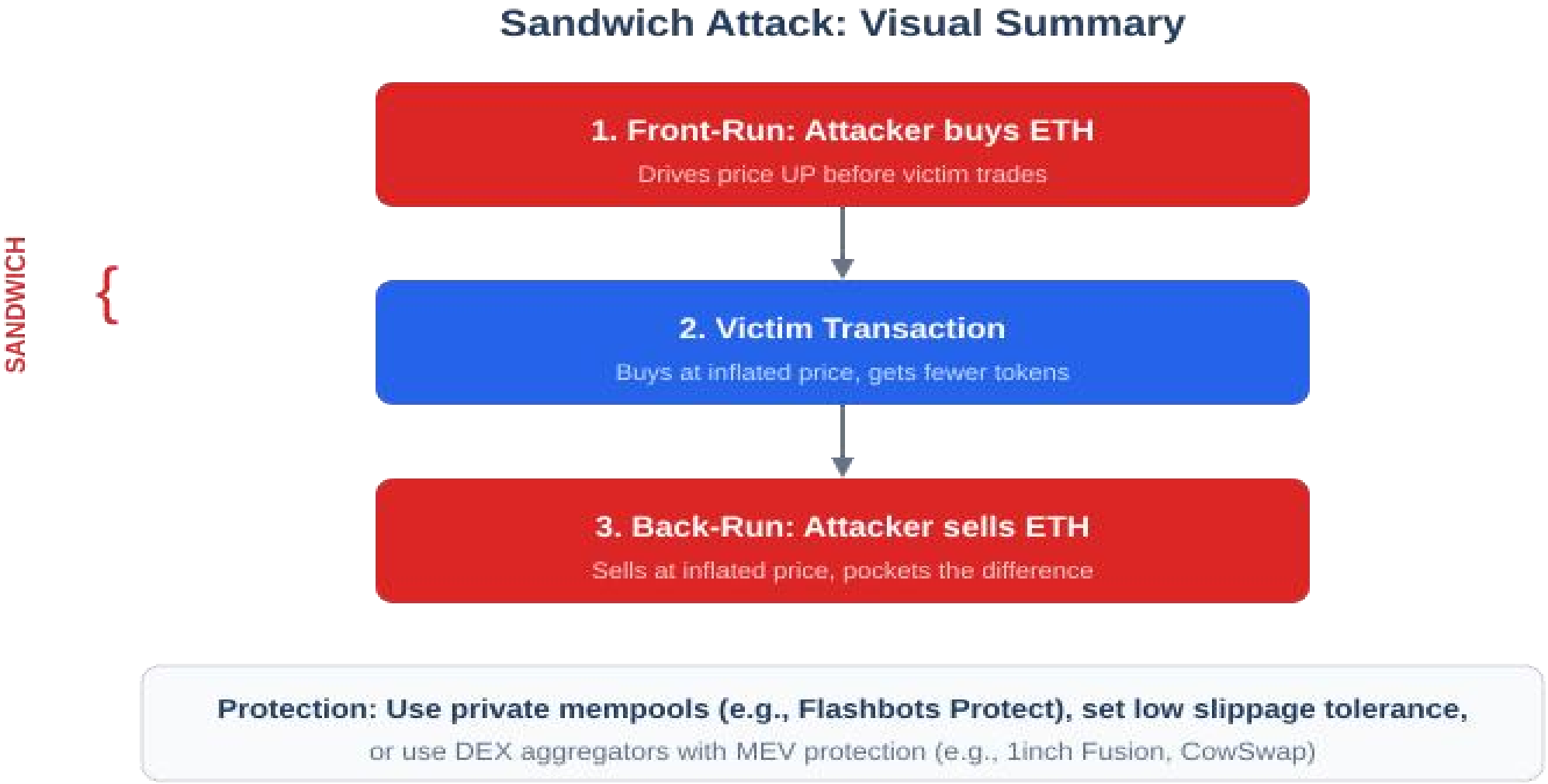
A sandwich attacker may submit a front-running transaction and a back-running transaction to get 2 ETH as revenue from the victim transaction.

Sandwich Attack Example

After the third transaction from the trader, the reserves in the liquidity pool becomes:

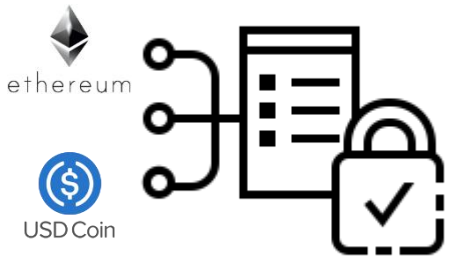


Protecting Against Sandwich Attacks

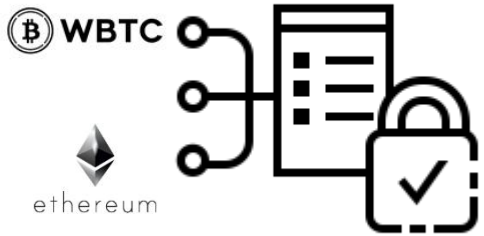


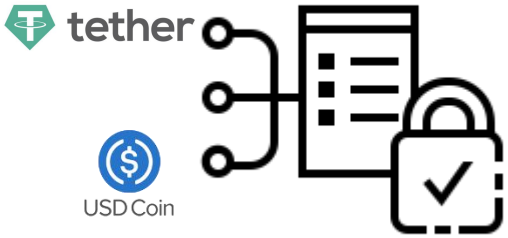
Protection: Use private mempools (e.g., Flashbots Protect), set low slippage tolerance,
or use DEX aggregators with MEV protection (e.g., 1inch Fusion, CowSwap)

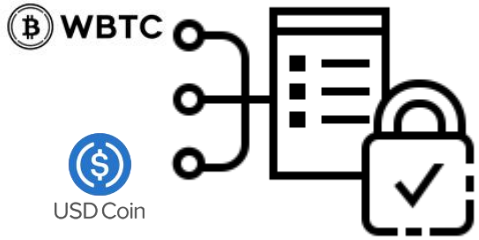
Decentralized Exchange Markets

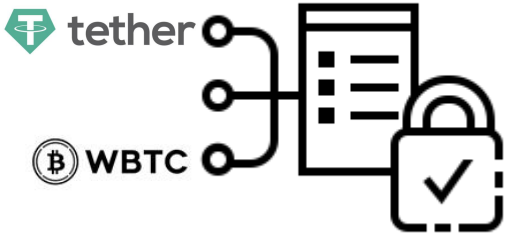






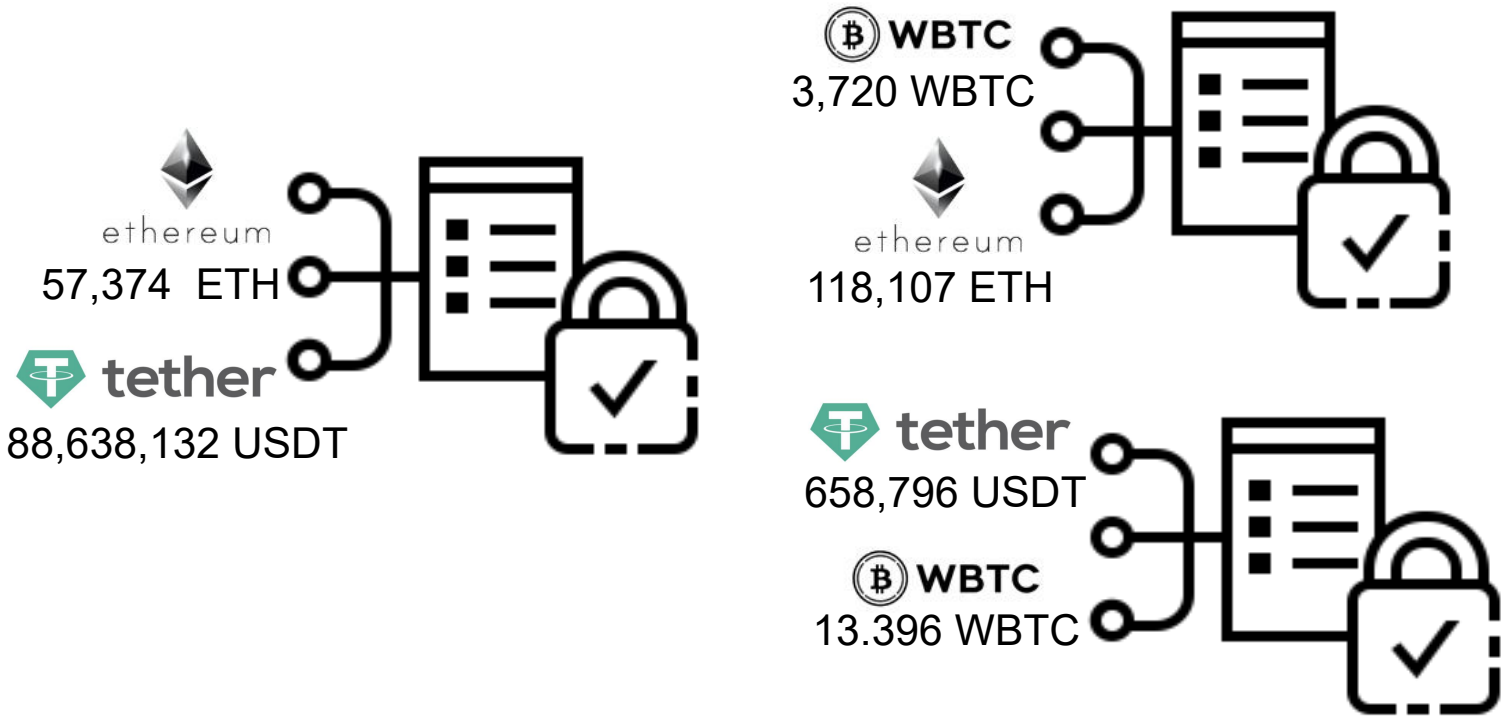




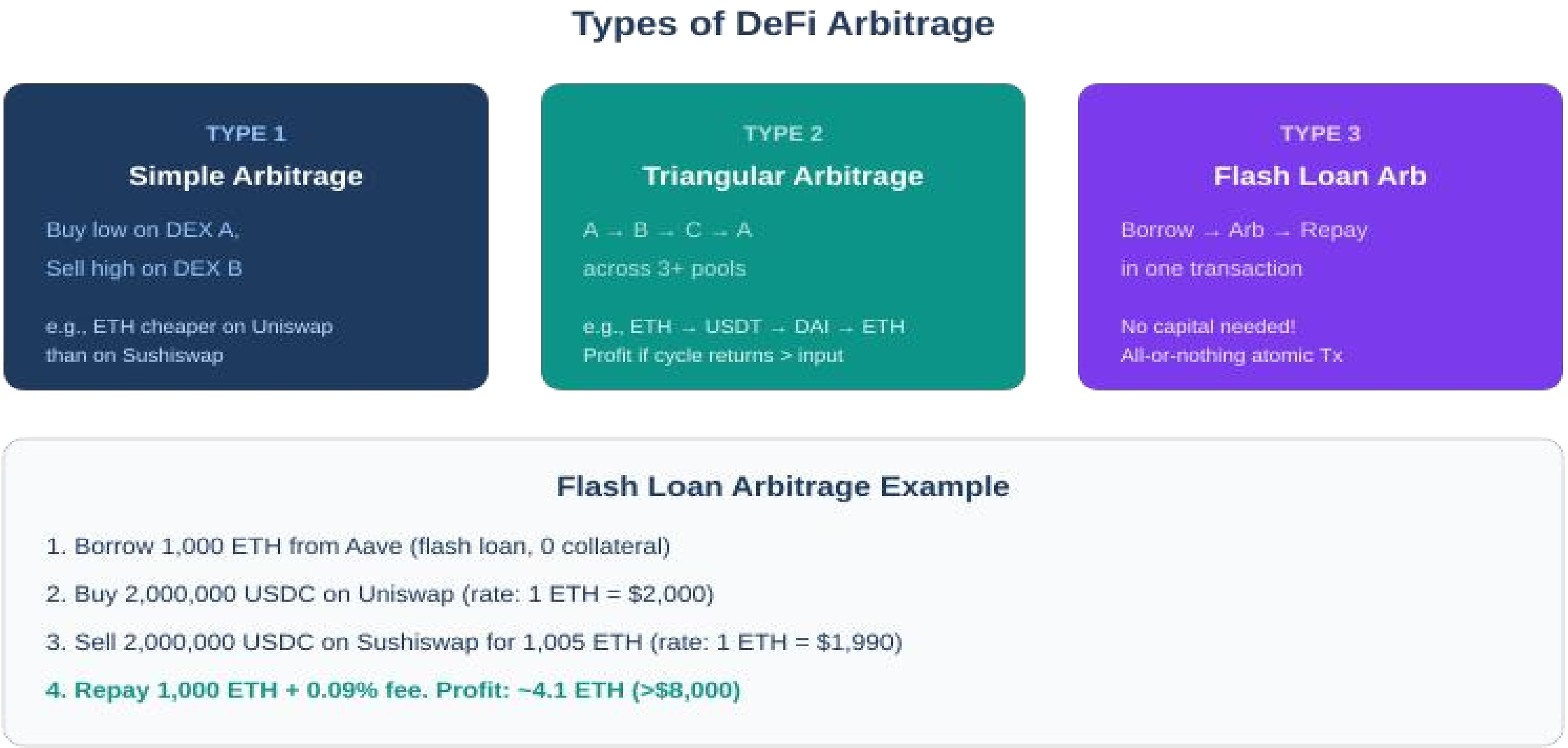


Unbalanced Exchange Rates

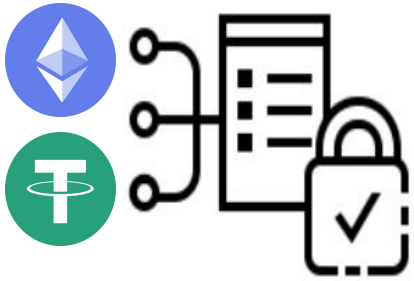
- What is arbitrage?
  - Riskless profit obtained by exploiting price differences.
- What happens after arbitrage?
  - Both pools reach same price – arbitrage no longer profitable after this.



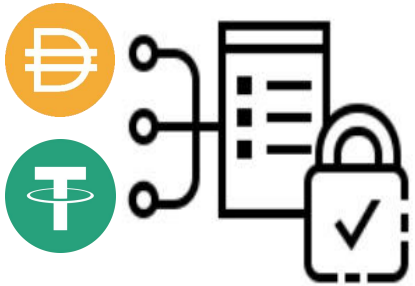
Types of DeFi Arbitrage



Arbitrage Example



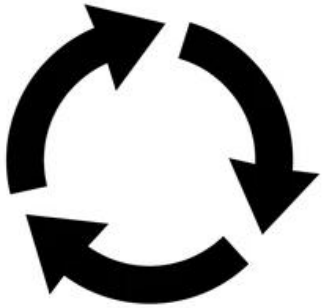
80 ETH
100 USDT

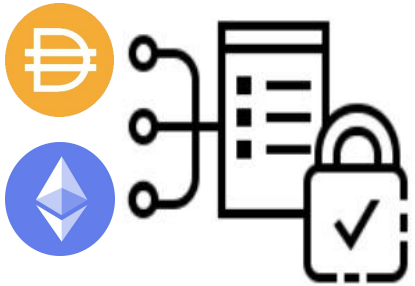


900 DAI
100 USDT



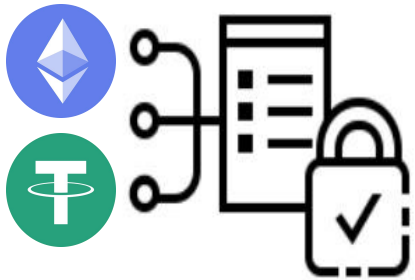
Arbitrager



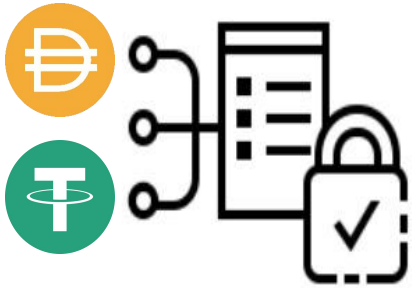


800 DAI
150 ETH

Arbitrage Example



80 ETH
100 USDT

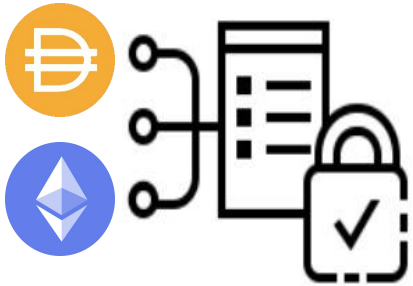


900 DAI
100 USDT



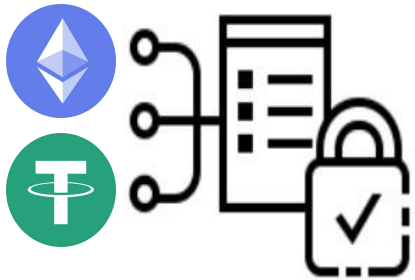
10 ETH

Arbitrager



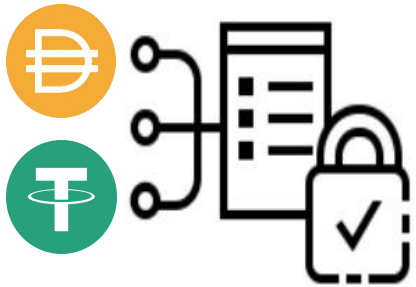
800 DAI
150 ETH

Arbitrage Example



90 ETH
89 USDT

11 USDT

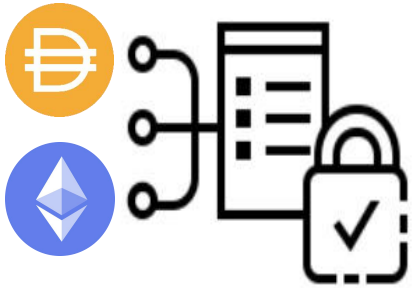


900 DAI
100 USDT

10 ETH

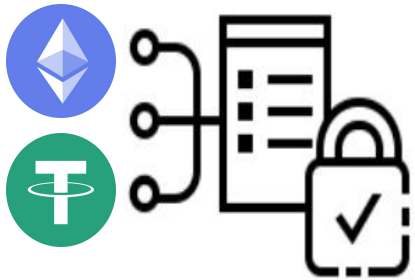


Arbitrager



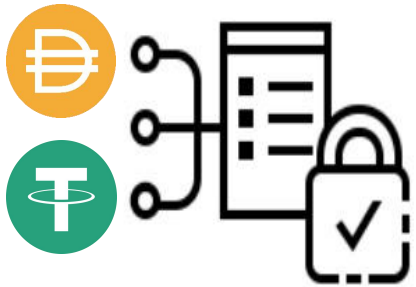
800 DAI
150 ETH

Arbitrage Example



90 ETH
89 USDT

11 USDT



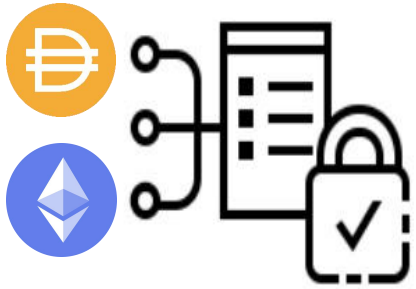
811 DAI
111 USDT

89 DAI

10 ETH

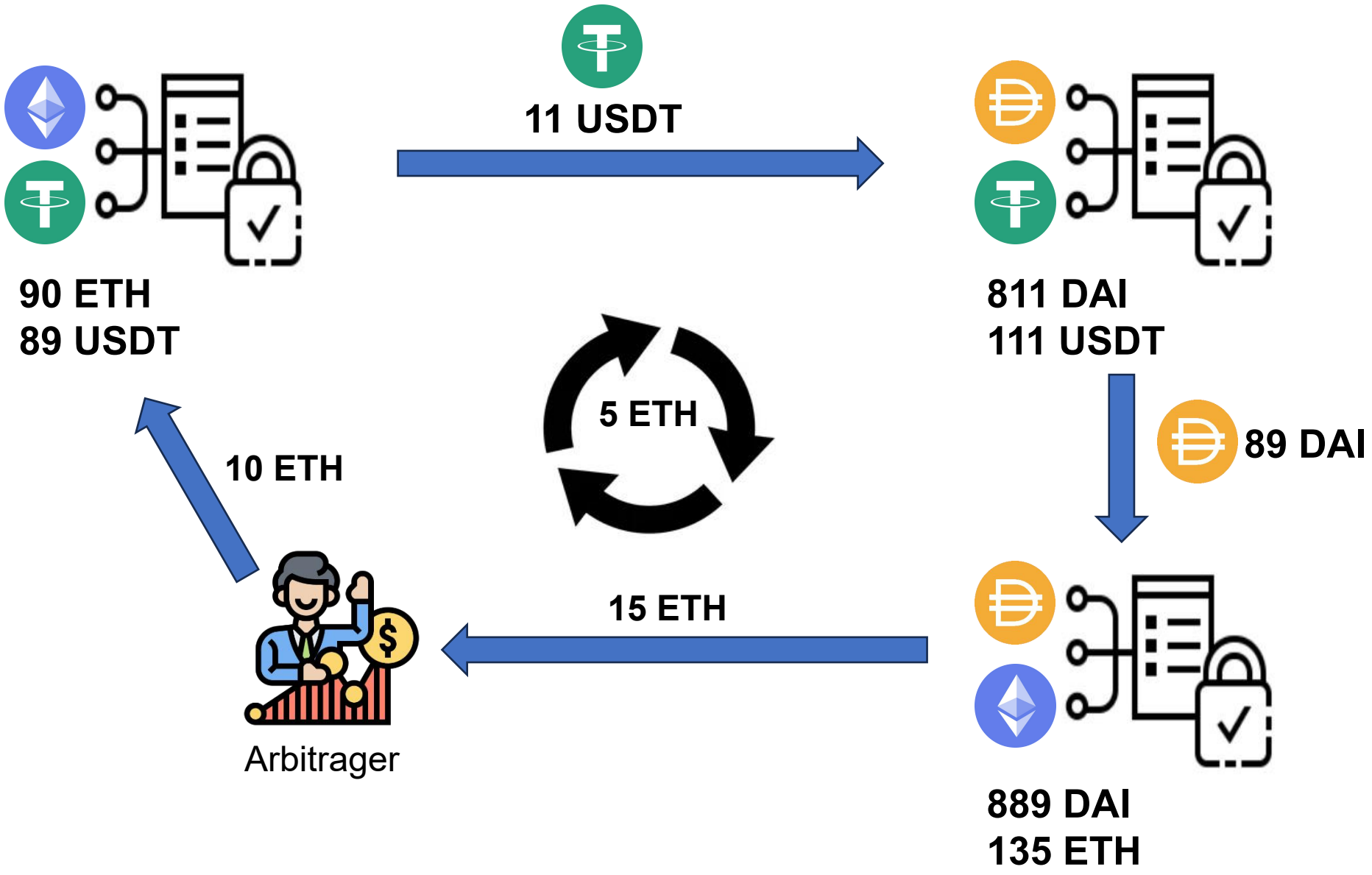


Arbitrager

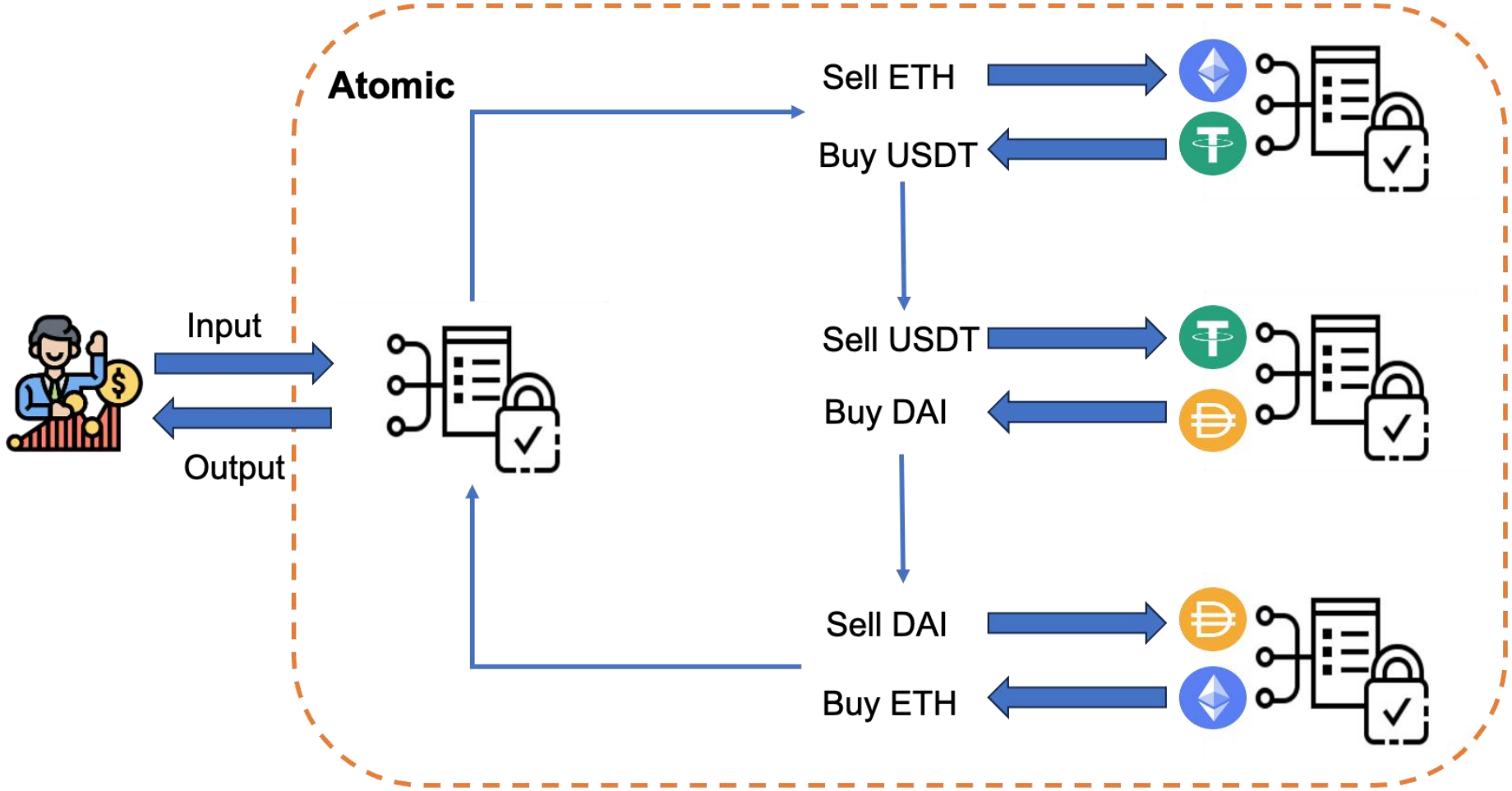


800 DAI
150 ETH

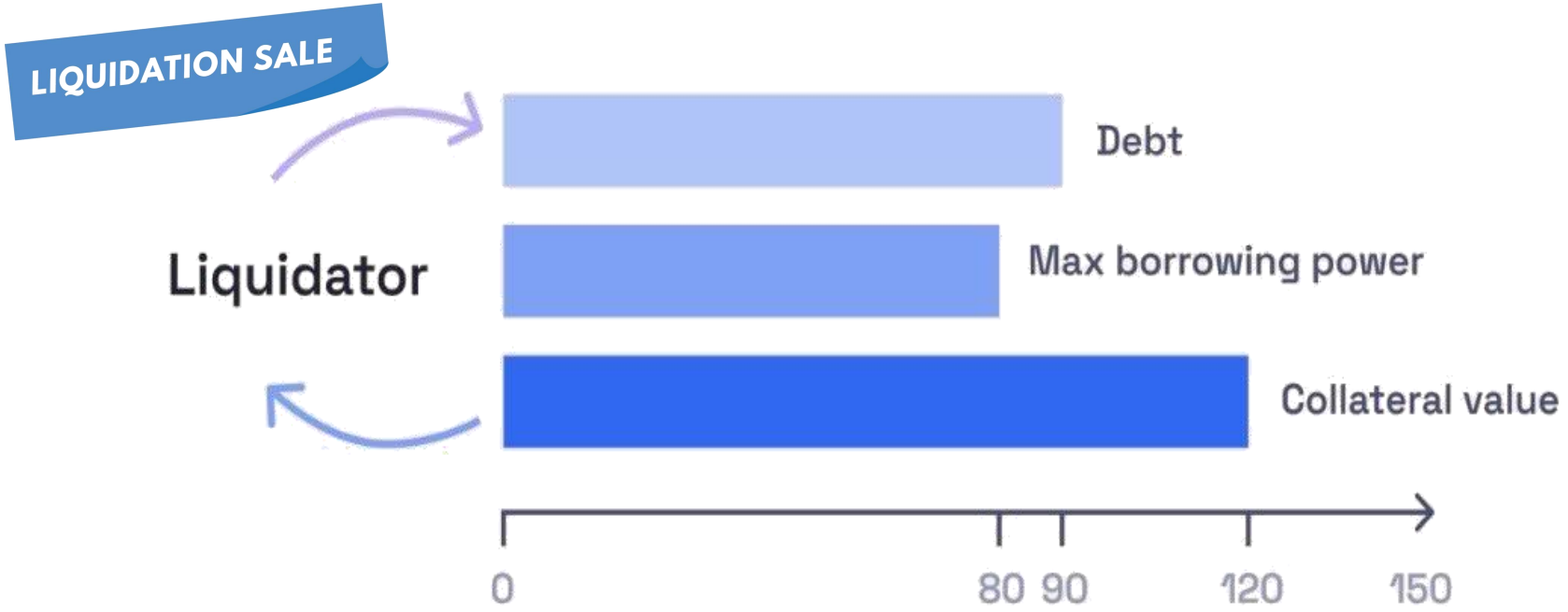
Arbitrage Example



Implementation



Liquidation

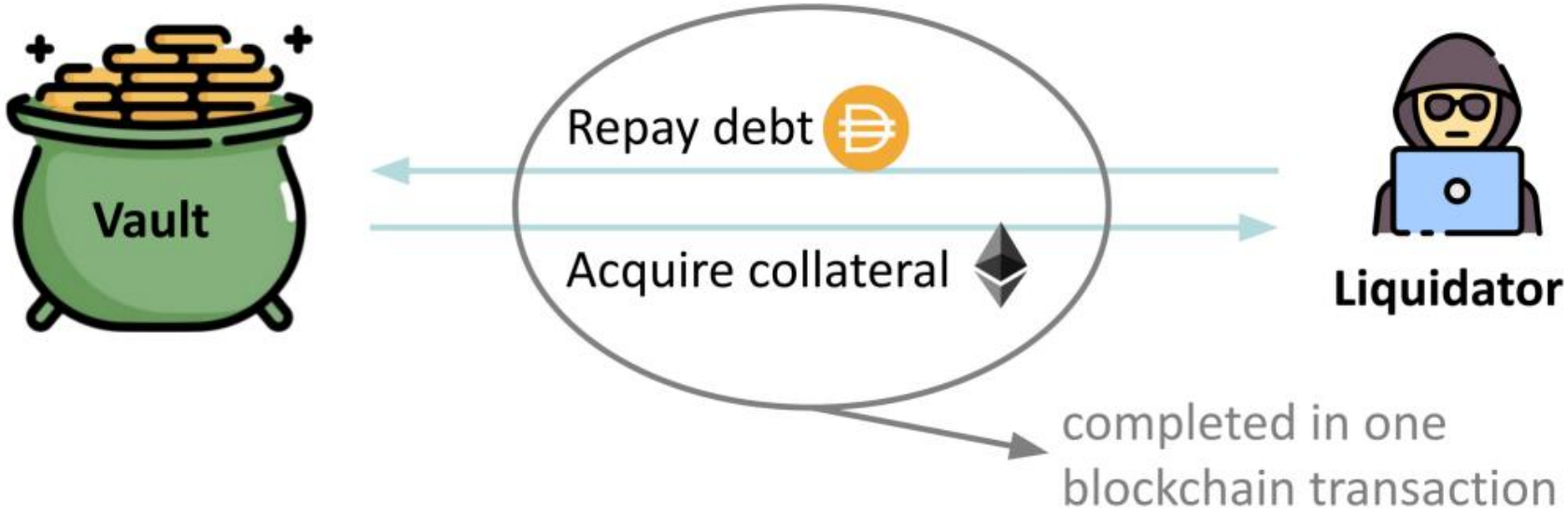


- The debt can get liquidated if the value of the collateral deposited gets too low compared to the value of the debt borrowed.
- Liquidation is the mechanism by which external actors repay the debt of borrowers on their behalf when they become insolvent.

Liquidation

- Liquidation == Selling collateral from the borrower
- Liquidation Spread
  - Bonus, or discount for liquidator
  - Fixed spread, or variable (auction based)

Fixed Spread Liquidation



- Repays the debts of a borrowing position
- Acquires the collateral at a discounted price from the position in return
- Typical discounts are e.g.,5-15%in Aave

Liquidation: Why It Matters for DeFi

Without Liquidation

Alice deposits 1 ETH ($2,000)
Borrows 1,000 DAI
ETH crashes to $500

Problem:
Collateral ($500) < Debt ($1,000)
Alice has no incentive to repay!
Protocol holds a loss of $500

Result:
Bad debt accumulates
DAI loses its peg
Protocol becomes insolvent

With Liquidation

Alice deposits 1 ETH ($2,000)
Borrows 1,000 DAI (CR = 200%)
ETH drops to $1,300 (CR = 130%)

Liquidation triggered (CR < 150%):
Bob (liquidator) repays 1,000 DAI
Bob receives ETH at 5% discount
Bob pays ~$1,235 for $1,300 ETH

Result:
Protocol debt is repaid
Bob profits ~$65 (liquidation bonus)
System remains solvent!

Fixed Spread Liquidation

Suppose:

Required: Liquidity ratio (LR) = $\frac{\text{Collateral value}}{Maxmium\ Collateralized\ Debt\ Position\ (CDP)}$ = 170%



Alice wants to use 1 ETH to borrow DAI

Date	2025.01
Collateral	1 ETH
Collateral value	1700 DAI
Maximum CDP	1000 DAI
Actual CDP	500 DAI
Collateral ratio	340%



Oracle

(2025.01)
1 ETH= 1700 DAI

Collateral ratio (CR) = $\frac{\text{Collateral value}}{Actual\ CDP}$

If CR < LR, Alice's loan will be liquidated

Fixed Spread Liquidation

Suppose:

Required: Liquidity ratio (LR) = $\frac{\text{Collateral value}}{Maxmium\ Collateralized\ Debt\ Position\ (CDP)}$ = 170%



Alice wants to use 1 ETH to borrow DAI

Date	2025.01
Collateral	1 ETH
Collateral value	1700 DAI
Maximum CDP	1000 DAI
Actual CDP	500 DAI
Collateral ratio	340%



Oracle

(2025.01)
1 ETH= 1700 DAI

(2025.02)
1 ETH = 800 DAI

Collateral ratio (CR) = $\frac{\text{Collateral value}}{Actual\ CDP}$

If CR < LR, Alice's loan will be liquidated

Fixed Spread Liquidation

Suppose:

Required: Liquidity ratio (LR) = $\frac{\text{Collateral value}}{Maxmium\ Collateralized\ Debt\ Position\ (CDP)}$ = 170%



Alice wants to use 1 ETH to borrow DAI

Date	2025.02
Collateral	1 ETH
Collateral value	800 DAI
Maximum CDP	470.59 DAI
Actual CDP	500 DAI
Collateral ratio	160% < 170%



Oracle

(2025.02)
1 ETH = 800 DAI



Liquidation

If CR < LR, Alice's loan will be liquidated

Collateral ratio (CR) = $\frac{\text{Collateral value}}{Actual\ CDP}$ = 800/500 = 160%

Fixed Spread Liquidation

Suppose:

Required: Liquidity ratio (LR) = $\frac{\text{Collateral value}}{Maxmium\ Collateralized\ Debt\ Position\ (CDP)}$ = 170%



Alice wants to use 1 ETH to borrow DAI

Date	2025.02
Collateral	1 ETH
Collateral value	800 DAI
Maximum CDP	470.59 DAI
Actual CDP	500 DAI
Collateral ratio	160% < 170%

Repays the loan

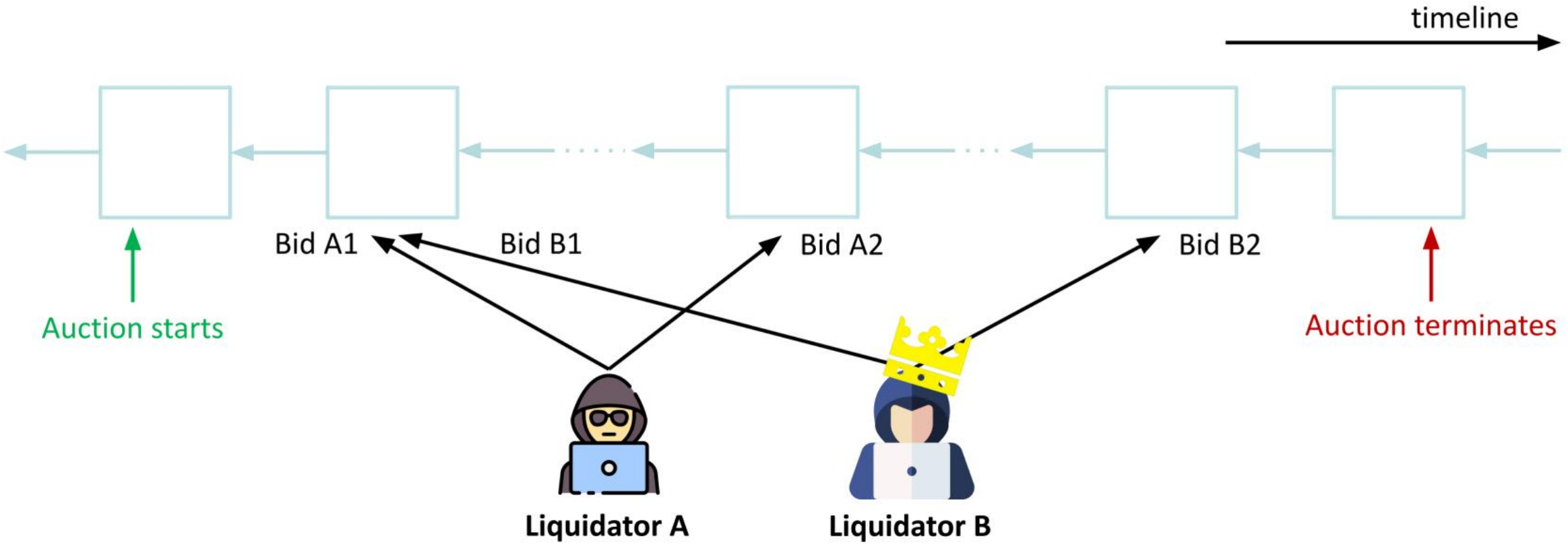


Liquidator

Liquidator could use 800*(1-5%) = **760 DAI** to buy **1 ETH (worth 800 DAI),** and immediately sells this 1 ETH at the market price of 800 DAI with 0.001 ETH gas fee.

Auction Liquidation

- Various liquidators bid over time until the auction terminates
- Requires multiple blockchain transactions



Auction Liquidation

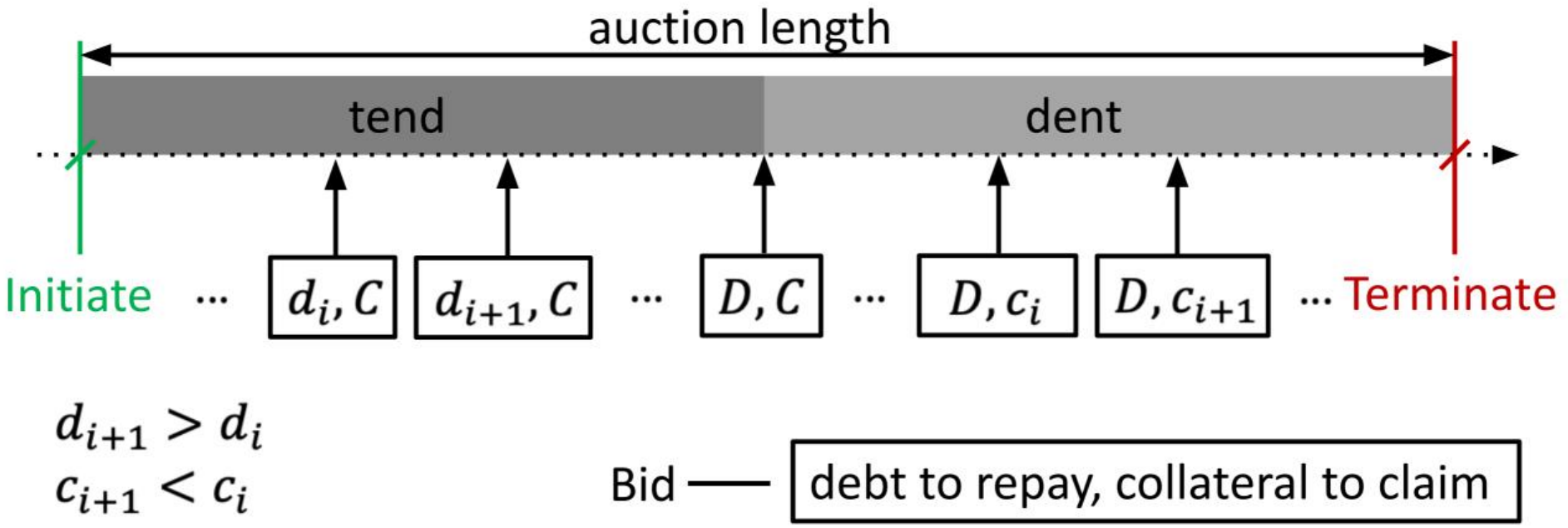
- **English Auction**
- bidders outbid each other increasingly

- **Dutch Auction**
- auction begins with a high asking price and the price lowers until the auction terminates

Auction Liquidation

MakerDAO tend-dent English auction

- A position with D debt and C collateral



History of MEV

- **Philip Daian** reported in 2019 that MEV appears in the priority transaction ordering of decentralized exchanges (DEXs), posing a systemic risk to the security of the consensus layer.
  (Reference: "Flash Boys 2.0: Frontrunning, Transaction Reordering, and Consensus Instability in Decentralized Exchanges")

- Recognizing the significance of MEV, Philip co-founded a company named **Flashbots** to offer solutions to MEV-related problems.



- Flashbots acknowledges that MEV is unavoidable and attempts to design a protocol that harnesses MEV to safeguard the system.

MEV: Historical Timeline

2019	**Flash Boys 2.0**	Philip Daian publishes groundbreaking paper on MEV
2020	**Flashbots Founded**	Dedicated org to research and mitigate MEV
2021	**MEV-Geth Launch**	First MEV auction system for miners (PoW)
2022	**The Merge + MEV-Boost**	Ethereum moves to PoS, Proposer-Builder Separation
2023+	**Builder Concentration**	3 builders control ~97% of blocks. New challenges emerge

Main Threat of MEV

- For miners: MEV profit > Transaction fee income & mining block reward

- Miners with high-performance servers can discover more MEV profit compared to others. They can share some of the MEV benefits with other miners, enticing them to join in rather than remaining as independent miners.

- Ultimately, the end game of this situation is that only a few miners with supercomputers will still be running the network.
  => Miner concentration and may transforms a **centralized** system!

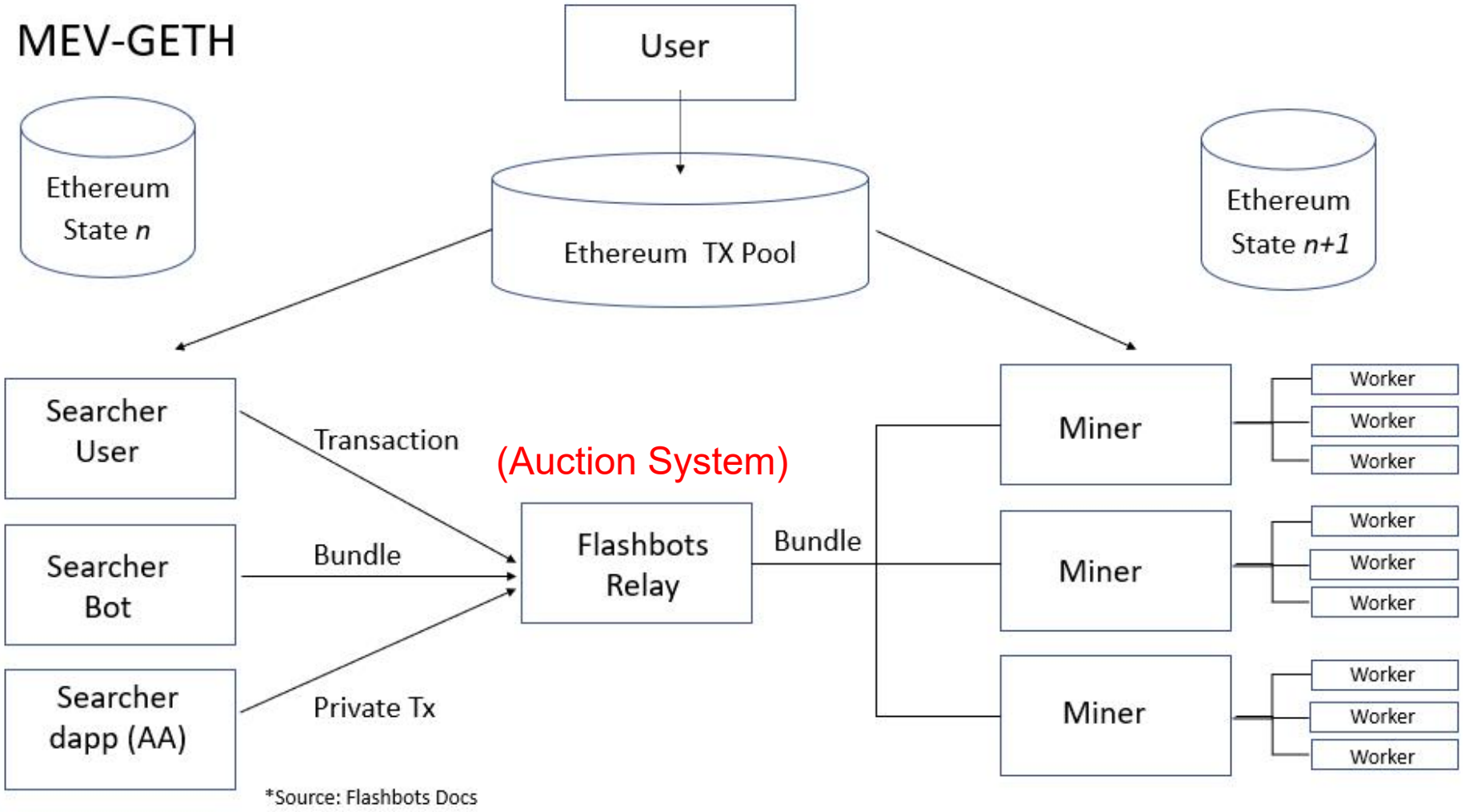
Solution provided by Flashbots

- MEV comes from the ability of miners to control transaction ordering.

- Flashbots attempts to democratize MEV by establishing an auction system for transaction ordering rights.
  => the creation of **MEV-Geth**

- The solution proposed by MEV-Geth is to introduce new actors known as **searchers** to share the workload of **finding MEV profit** with miners.

Architecture of MEV-Geth



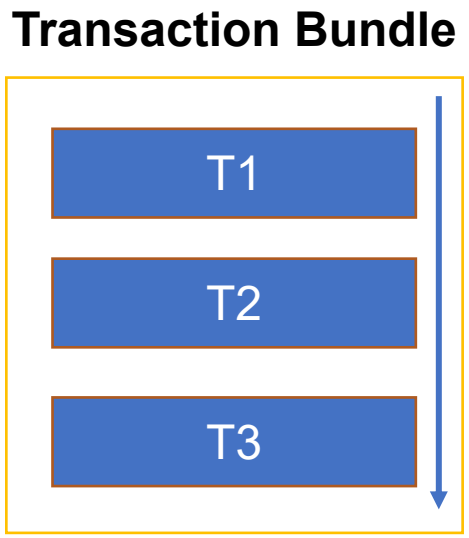
(Miner/Worker choose the bundles with highest prices)

Mechanism of MEV-Geth

- In MEV-Geth, searchers pay miners so that the bundle of transactions they created, including their MEV, can be added to the block via the auction system.

- A bundle of transactions includes several transactions with ordering determined by searchers. Miners need to follow this ordering when include this bundle into a block.

- For example, a searcher creates both frontrun and backrun transactions in a sandwich attack to obtain MEV. Then, the searcher shares a portion of the MEV profit with the miner (by offering a price through the auction system) so that the miner will include these transactions in the block.
  Then, miners select several bundles of transactions with the highest prices into the block, and earn the rewards provided by searchers when they successfully mine a block.



Benefit of MEV-Geth

1. Prevent miner concentration: Make it possible for every miner to earn MEV profits from searchers. Miners only need to calculate the hash of the block.

2. Achieve fair MEV profit distribution: In MEV-Geth, anyone can act as a searcher. This means that not only miners but also any individual has an opportunity to earn MEV profit if they find the chance.

3. Eliminate gas price auctions in the public mempool
    - Instead, create an off-chain market where searchers can compete for the position of their bundles in a block

New Problem of MEV-Geth After Upgraded to PoS

- After The Merge, Ethereum changed its consensus protocol from Proof of Work (PoW) to Proof of Stake (PoS). However, new problems emerged.

- In PoS, the wealthy can own more proposers (validators in PoS, not miners as in PoW) than others, enabling them to control which transaction bundles are included in blocks.
  => Proposer concentration and may transforms a **centralized** system again!

- To address this issue, Flashbots proposed the **Proposer-Builder Separation (PBS)** solution and created **MEV-Boost**.
  It introduces new participants called **builders** to share the workload of selecting transaction bundles, determining their order and assembling them into a complete block for the proposer.

Evolution: MEV-Geth → MEV-Boost

MEV-Geth vs MEV-Boost Evolution

MEV-Geth (PoW Era)

- Searchers → Flashbots Relay → Miners
- Single relay (Flashbots only)
- Miners select bundles + build blocks
- Miners have dual role (build + propose)

Problem: Single point of failure

Flashbots relay = centralization risk

Problem: Miner concentration

Powerful miners dominate MEV extraction

MEV-Boost (PoS Era)

- Searchers → Builders → Relays → Proposers
- Multiple relays (open market)
- Builders select bundles + build blocks
- Proposer-Builder Separation (PBS)

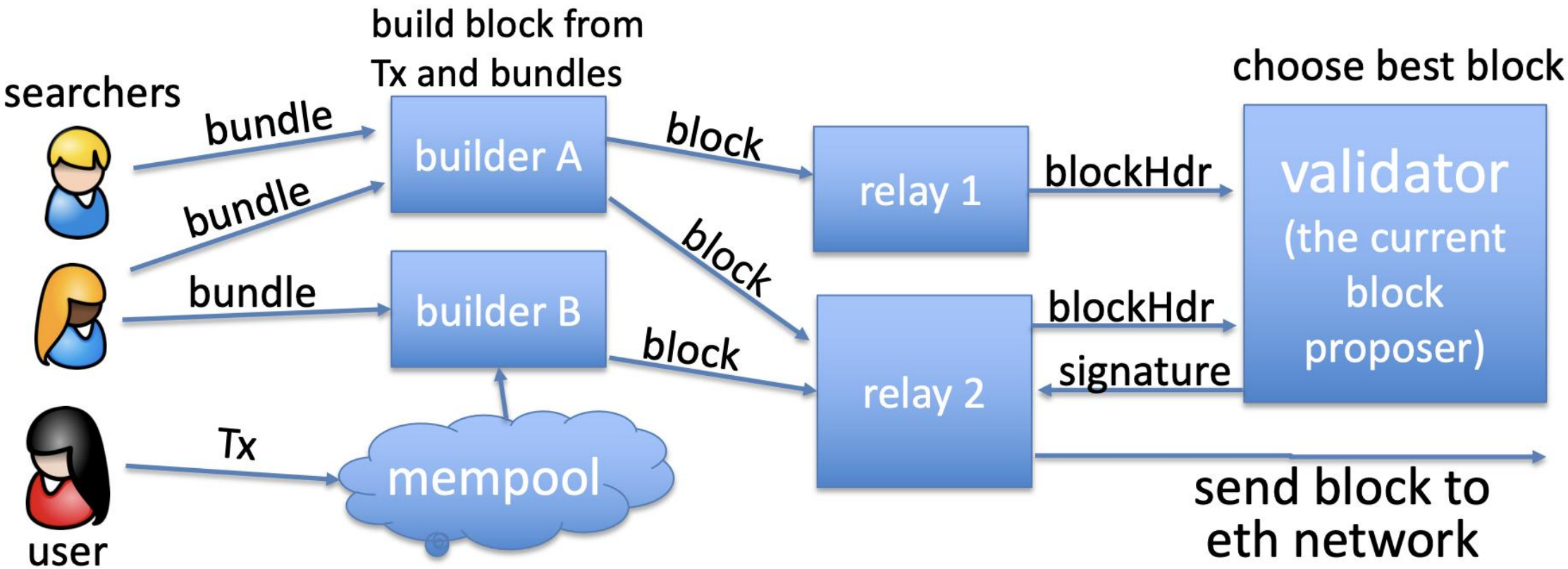
Solution: Multiple competing relays

Decentralized relay market

Solution: Proposers just validate

Any computer can be a proposer

Architecture of MEV-Boost (PBS Solution)



New Benefits of MEV-Boost

1. Fair Block Building Ability: Builders operate off-chain and face no limitations, so anyone can become a builder. All possible blocks are created by builders and sent to the relays (auction systems). These blocks compete to be the best block, which makes the process fairer.

2. More Decentralization of Proposers: In MEV-Boost, proposers only need to collect information about the best block from relays and broadcast it. Any computer can perform this task, enhancing the decentralization of proposers.

3. Relay Democratization: Unlike in MEV-Geth, where only Flashbots functioned as the relay. In MEV-Boost, anyone can act as a relay.

Characters in MEV-Boost

Builder: collects bundles and Transactions, builds a block (≈300 bundles/block)

- includes a MEV offer to validator (feeRecipient)

Relay: collects blocks, chooses block with max MEV offer

- sends block header (and MEV offer) to block proposer
- Can't expose Transactions in block to proposer (proposer could steal Transactions)

Proposer: chooses best offer and signs header with its staking key

⇒ Then Relay sends block to network, making it public

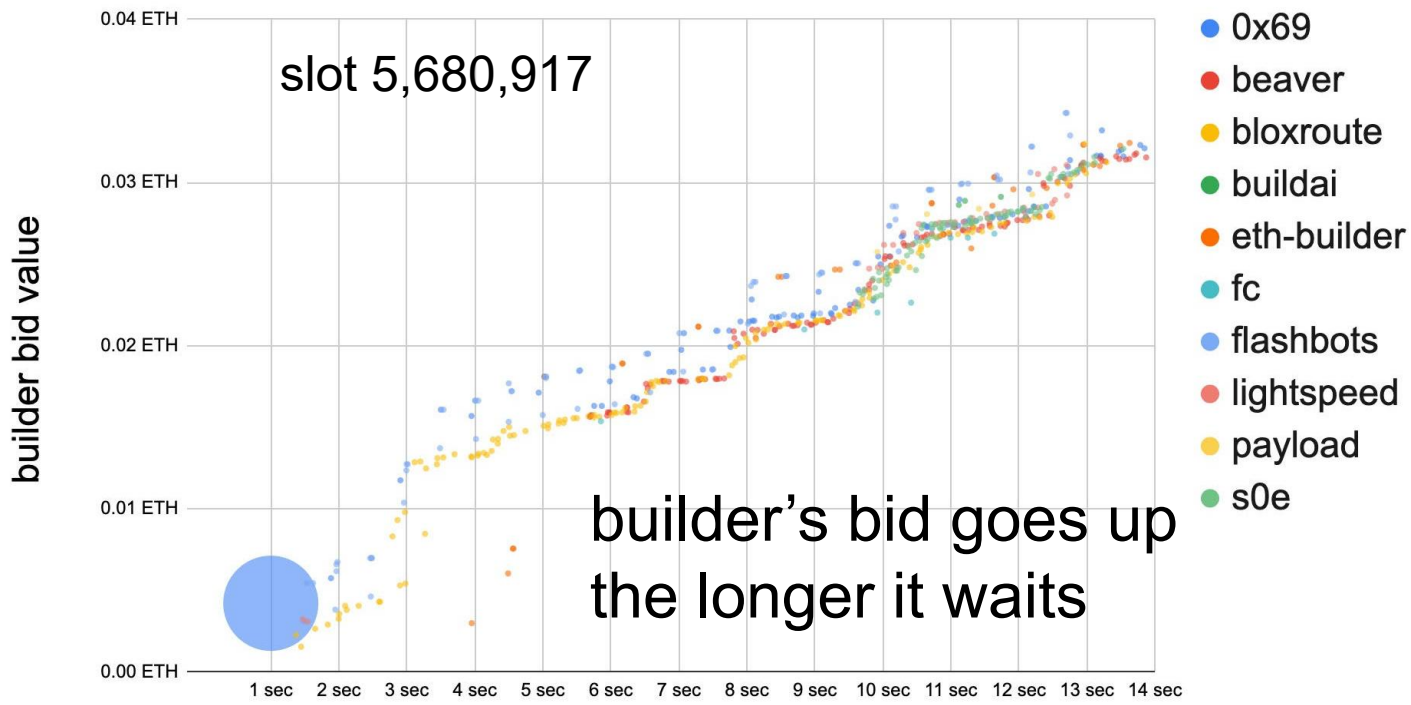
⇒ Now, proposer cannot steal MEV (would be exposed to slashing)

Many Block Options Per Slot

A relay might receive 500 blocks per slot from builders

Each builder might send 20 blocks to relay for one slot

Why? The longer builder waits the more MEV opportunities …



Operating Relays

Flashbots: Filters out OFAC sanctioned addresses, aims to maximize validator payout

BloXroute: no censorship, aims to maximize validator payout

UltraSound: not for profit, non censoring

Relay	Payloads	Percent
relay.ultrasound.money	29,208	30.95 %
bloxroute.max-profit.blxrbdn.com	23,788	25.21 %
bloxroute.regulated.blxrbdn.com	21,379	22.66 %
titanrelay.xyz	10,855	11.50 %
agnostic-relay.net	3,651	3.87 %
boost-relay.flashbots.net	3,301	3.50 %
aestus.live	2,121	2.25 %
relay.edennetwork.io	63	0.07 %

https://www.relayscan.io/overview?t=7d

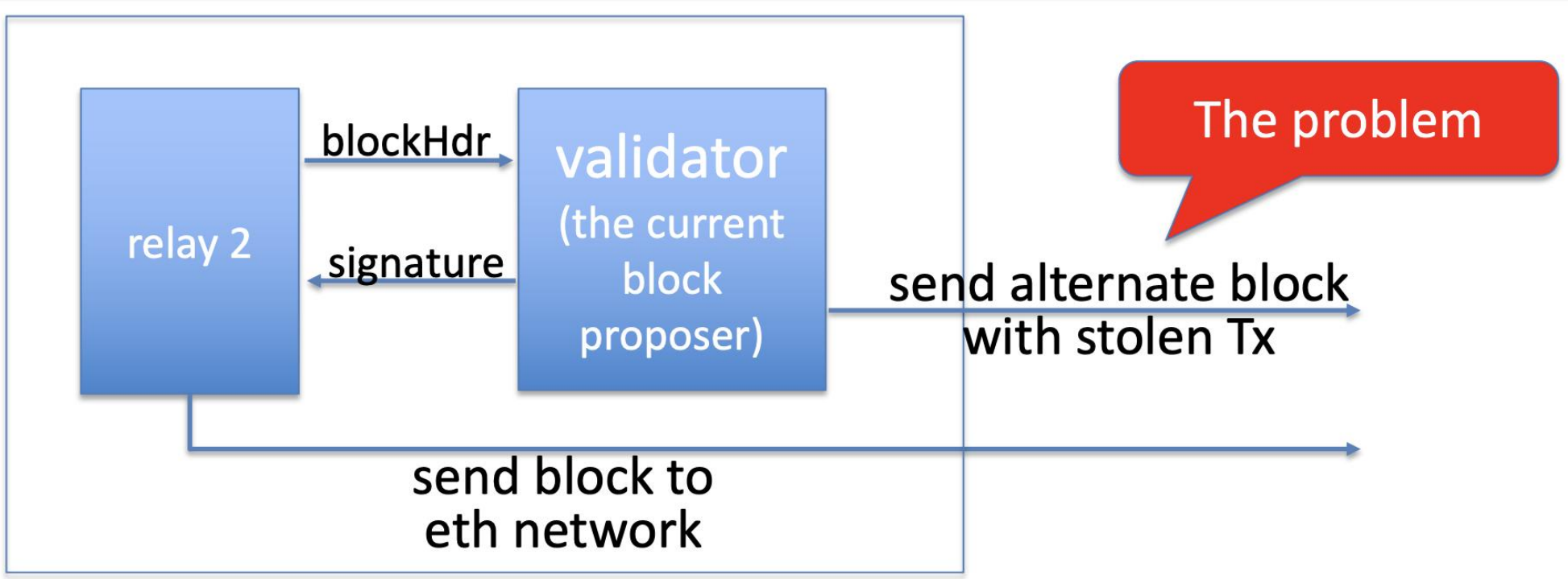
An Example: Flashbots Relay

Recently Delivered Payloads

fee to validator

Epoch	Slot	Block number	Value (ETH)	Num tx
165,046	5,281,503	16,115,184	0.0759673152	186
165,046	5,281,501	16,115,182	0.05098935853	142
165,046	5,281,499	16,115,180	0.1902791095	167
165,046	5,281,498	16,115,179	0.103438972	295
165,046	5,281,496	16,115,177	0.07159735143	199
165,046	5,281,495	16,115,176	0.04034671944	125

The Race Problem



Block proposer will be slashed (why?) ⇒ Lose 1 ETH

… but can gain much more in stolen MEV.

Are We Done? Not Quite …

(Titan, beaverbuild, rsync-builder)

Builder concentration: 3 builders build around 97% of all blocks !!

- Clear centralization in the builder market
- Enables censorship by builders

Builder (extra_data)	Blocks	Percent
beaverbuild.org	19,834	43.45 %
Titan (titanbuilder.xyz)	19,030	41.69 %
rsync–builder.xyz ⓘ	5,380	11.79 %

Proposers hold all the power (first price auction among builders)

⇒ Most MEV profits flow to block proposers

MEV-boost is not designed for cross-chain MEV

- For cross-chain arbitrage, no atomicity guarantee for bundle

Part 2 Recap: Key Takeaways



MEV = profit from including, excluding, or reordering transactions in blocks



Three MEV types: sandwich attacks (harmful), arbitrage (beneficial), liquidation (necessary)



MEV-Geth democratized MEV extraction via auction system for miners



MEV-Boost introduced PBS: separating block building from block proposing



Current challenge: builder concentration — 3 builders build ~97% of blocks



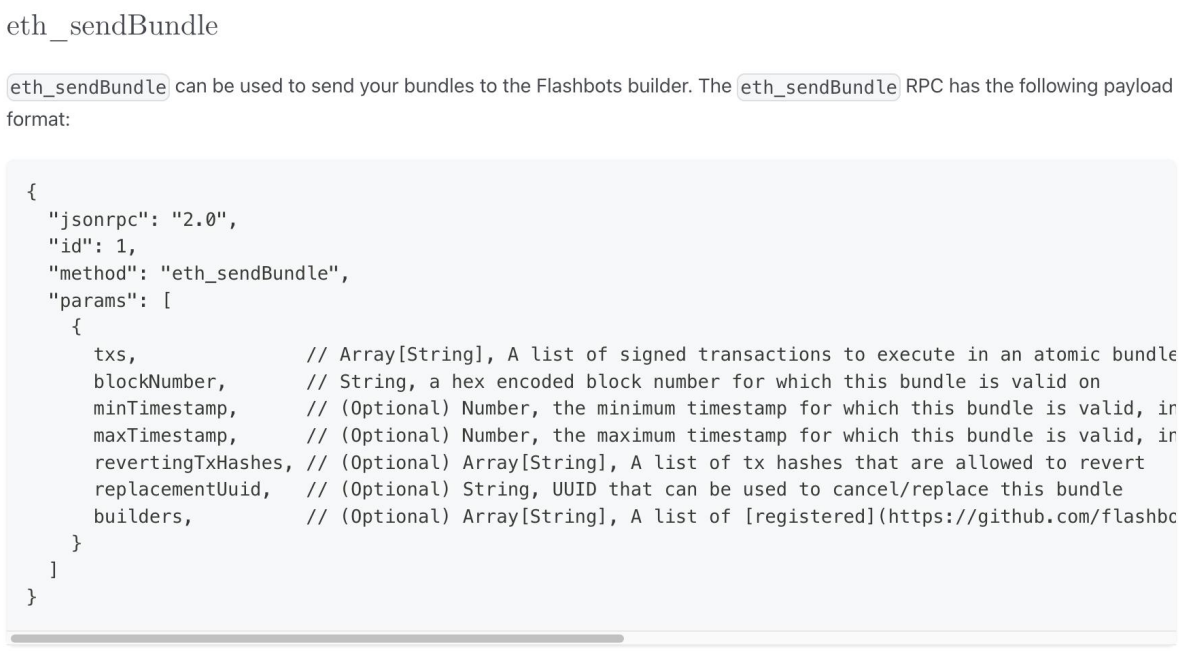
Protection tools: Flashbots Protect, CowSwap, 1inch Fusion, low slippage settings

Workshop

1. Method of Sending Transaction Bundles

- In MEV-Boost, there is no relay between searchers and builders. Searchers must send transaction bundles directly to builders via the API.

- For example, Flashbots provides an API "eth_sendBundle" to receive transaction bundles. https://docs.flashbots.net/flashbots-auction/advanced/rpc-endpoint#eth_sendbundle



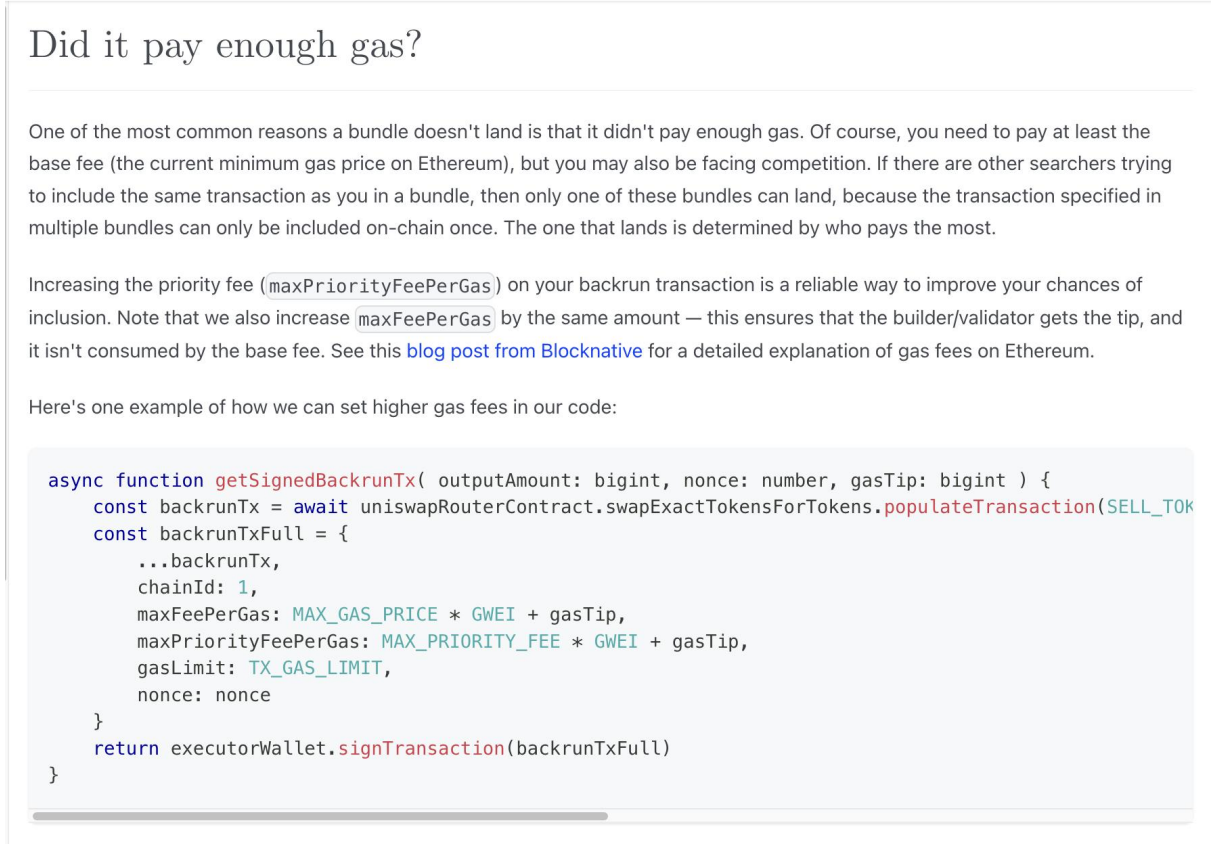
eth_sendBundle

`eth_sendBundle` can be used to send your bundles to the Flashbots builder. The `eth_sendBundle` RPC has the following payload format:

```
{
  "jsonrpc": "2.0",
  "id": 1,
  "method": "eth_sendBundle",
  "params": [
    {
      txs,               // Array[String], A list of signed transactions to execute in an atomic bundle
      blockNumber,       // String, a hex encoded block number for which this bundle is valid on
      minTimestamp,      // (Optional) Number, the minimum timestamp for which this bundle is valid, ir
      maxTimestamp,      // (Optional) Number, the maximum timestamp for which this bundle is valid, ir
      revertingTxHashes, // (Optional) Array[String], A list of tx hashes that are allowed to revert
      replacementUuid,   // (Optional) String, UUID that can be used to cancel/replace this bundle
      builders,          // (Optional) Array[String], A list of [registered](https://github.com/flashbc
    }
  ]
}
```

2. How do searchers pay builders for the bundle?

- Following the documentation provided by Flashbots, searchers pay via the transaction fee, the value of which is determined by the gas price. Therefore, if searchers want to win the auction, they have to offer a higher priority fee (gas price).



Did it pay enough gas?

One of the most common reasons a bundle doesn't land is that it didn't pay enough gas. Of course, you need to pay at least the base fee (the current minimum gas price on Ethereum), but you may also be facing competition. If there are other searchers trying to include the same transaction as you in a bundle, then only one of these bundles can land, because the transaction specified in multiple bundles can only be included on-chain once. The one that lands is determined by who pays the most.

Increasing the priority fee (`maxPriorityFeePerGas`) on your backrun transaction is a reliable way to improve your chances of inclusion. Note that we also increase `maxFeePerGas` by the same amount — this ensures that the builder/validator gets the tip, and it isn't consumed by the base fee. See this blog post from Blocknative for a detailed explanation of gas fees on Ethereum.

Here's one example of how we can set higher gas fees in our code:

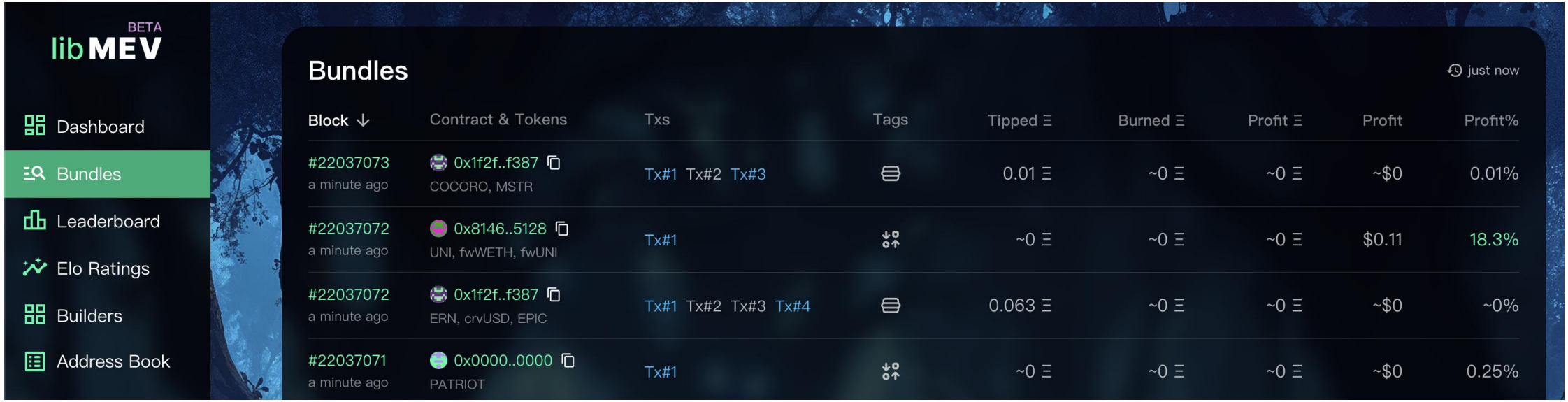
```
async function getSignedBackrunTx( outputAmount: bigint, nonce: number, gasTip: bigint ) {
    const backrunTx = await uniswapRouterContract.swapExactTokensForTokens.populateTransaction(SELL_TOK
    const backrunTxFull = {
        ...backrunTx,
        chainId: 1,
        maxFeePerGas: MAX_GAS_PRICE * GWEI + gasTip,
        maxPriorityFeePerGas: MAX_PRIORITY_FEE * GWEI + gasTip,
        gasLimit: TX_GAS_LIMIT,
        nonce: nonce
    }
    return executorWallet.signTransaction(backrunTxFull)
}
```

https://docs.flashbots.net/flashbots-mev-share/searchers/debugging#did-it-pay-enough-gas

3. A Tool for Checking Latest Bundles

- Libmev.com offers a website for checking transaction bundles that appear in blocks. It can also indicate the type of MEV included in each bundle.

  https://libmev.com/bundles/



4. Method of Submitting Blocks from Builders

- Flashbots provides an API that allows builders to submit their constructed blocks to the Flashbots Relay.

External Builders

External builders can submit blocks to Mainnet, Goerli and Sepolia Flashbots relays. The table below outlines Builder API methods available on each network.

Relay Block Submission Endpoints by Network

		Mainnet	Goerli	Sepolia
`getValidators`	GET Request - Returns an array of validator registrations with assigned duties in the current and next epoch	Mainnet	Goerli	Sepolia
`submitBlock`	POST Request - submits a block to the relay	Mainnet	Goerli	Sepolia

- See also the Relay API documentation - Block Builder API for more details on the API and payloads.
- The example Flashbots builder implementation is a good external builder reference, and is currently used in production by several builders.

https://docs.flashbots.net/flashbots-mev-boost/block-builders#external-builders

https://flashbots.github.io/relay-specs/#/

5. Method of Receiving Block Information as Proposer

- Flashbots offers several APIs that enable proposers to retrieve headers and payloads from the Flashbots Relay in a step-by-step manner.

Proposer API

Standard builder spec APIs

- registerValidator: `POST /eth/v1/builder/validators`
- getHeader: `GET /eth/v1/builder/header/{slot}/{parent_hash}/{pubkey}` - Get an execution payload header.
- submitBlindedBlock: `POST /eth/v1/builder/blinded_blocks` - Submit a signed blinded block and get unblinded execution payload.
- status: `GET /eth/v1/builder/status`

https://docs.flashbots.net/flashbots-mev-boost/relay#proposer-api

6. Several Useful Tools for Analyzing MEV

1. **ZeroMEV**: A website provides all MEVs that appear in a single block and estimates the loss associated with each MEV.
   https://www.zeromev.org/block?num=19021324

2. **RelayScan**: A website shows the concentration of relays and builders over a period of time.
   https://www.relayscan.io/overview?t=24h

3. **MEV-Boost Pics**: An overall dashboard that displays information about MEV.
   https://mevboost.pics/

4. For more tools, Flashbots provides a page to introduce useful resources.
   https://docs.flashbots.net/community-tools